Acta Ophthalmologica

Acta Ophthalmol. 2014: 92 (Suppl. 254): 1–23
doi: 10.1111/aos.12582

Abstracts of the Nordic Congress of Ophthalmology, 20–23 August 2014, Stockholm, Sweden

FP-CAT-031

Two-year follow-up of PCO development and glistenings for two 1-piece hydrophobic acrylic IOLs – an intraindividual comparison

Björn Johansson
Department of Ophthalmology and Department of Clinical and Experimental Medicine, Linköping University, Linköping, Sweden

Purpose: To compare two hydrophobic intraocular lenses regarding development of PCO and glistenings during two years after implantation.
Methods: To this intraindividual, prospective randomised study participants with bilateral cataract planned for immediate sequential bilateral cataract surgery (ISBCS) were invited. Exclusion criteria were other visually disturbing conditions, previous intraocular surgery, or any complication damaging lens capsule or zonulae. 50 patients participated. The first operated eye was randomised to receive either an AcrySof IQ intraocular lens or a Tecnis One-Piece intraocular lens. Visual acuity and intraocular pressure measurements were performed as well as a slitlamp examination and photography 1 week and 2 years after surgery. PCO was measured using POCOman software. Glistenings in the IOL optic were measured semi quantitatively with a 4 graded scale.
Results: At 2-year follow-up 44 patients returned for examination. 6 patients died before 2-year follow-up. 1 eye (with a Tecnis IOL) has been treated with Nd-YAG-laser capsulotomy. Mean POCOman area score was 11.0% for Tecnis eyes and 12.5% for AcrySof eyes, and mean severity score 0.15 for both groups. Glistenings appeared in 2 Tecnis IOLs (Grade 1), in the other 42 Tecnis IOLs grade 0 glistenings were detected. 8 AcrySof IOLs display grade 0 glistenings, 3 AcrySof IOLs each were graded 1 and 2. 7 AcrySof IOLs showed grade 3 glistenings and 23 AcrySof had glistenings grade 4. Glistenings were not linked to any visual complaints.
Conclusion: Two years after cataract surgery, an intraindividual comparison between AcrySof IQ and Tecnis 1-piece IOLs reveal equal results regarding PCO development. Glistenings are readily observed in most AcrySof IOLs whereas only two of the 44 examined Tecnis IOLs displayed sparse glistenings. In the study population both IOLs provided equally good visual results and satisfying visual quality.

FP-CAT-048

Outcomes after phacoemulsification surgeries preoperatively categorised according to presumed surgical difficulty

Malin Westerlund, ThuBa Wihlmark, Johan Germundsson, Björn Johansson
Department of Ophthalmology and Department of Clinical and Experimental Medicine, Linköping University, Linköping, Sweden

Aim: To analyse outcomes after phacoemulsification surgery in cases preoperatively categorised after expected surgical difficulty.
Setting: Department of Ophthalmology at University Hospital.
Methods: Planned phacoemulsification surgery cases were preoperatively categorized according to expected surgical difficulty level. Criteria were: Category A: Healthy anterior segment except cataract. Category B: Submaximal pupil dilatation with or without exfoliations, vitrectomized eyes, high hyperopia/myopia, concurrent glaucoma, deep set eye. Category C: Cataract in an eye with contralateral eye blind, previous complicated surgery in other eye, dense white or black cataract, corneal endothelial dystrophy, uncertain cooperation e.g. senile dementia, tremor, small pupil, traumatic cataract, previous corneal transplantation, phacodonesis. 300 surgeries (100 by 3 surgeons each) were examined. Patient age, operation time, Effective Phaco Time and complicated surgical manoeuvres (mechanical pupil dilation, staining of anterior capsule, hooks stabilising capsulorhexis edge, capsular tension ring) were extracted from electronical patient records and analysed.
Results: One hundred and twenty-one cases were categorised as A, 132 as B and 47 cases as C. Mean operation times were 11.7 min (A), 13.8 min (B), and 17 min (C) respectively. Surgery time was longer than 20 min in 1 of 121 A cases (0.8%), 10 of 132 B cases (7.6%), and 10 of 47 C cases (21.3%). The number of complicated surgical manoeuvres was in group A 2 (0.017 per surgery), in group B 23 (0.17 per surgery), and in group C 28 (0.60 per surgery). In two cases (both B), zonular complication led to postoperative aphakia.
Conclusions: Categorising phacoemulsification surgery in three groups according to expected surgical difficulty lead to significant differences between the groups regarding surgery time and number of complicated surgical manoeuvres. This easily implemented approach is therefore helpful when scheduling phacoemulsification surgeries, facilitates optimisation of case mix in operation programmes for different surgeons, and may help to assign adequately competent surgeons to appropriate cases thereby increasing patient safety.

FP-CAT-060

Evaluation of an aberration corrected, diffractive and pupil independent, toric multifocal IOL (tMIOL)

Florian Kretz, Katharina Linz, Mary Safwat, Anna Fitting, Ramin Khormnia, Gerd Auffarth
International Vision Correction and Research Centre (IVCRC) & David J Apple Laboratory, Department of Ophthalmology, University of Heidelberg, Heidelberg, Germany

Purpose: Clinical evaluation of an aberration corrected, pure diffractive, pupil independent and toric MIOL.
Methods: In a prospective multicentre study a tMIOL (ZMT–, AMO, USA) was implanted in patients with a corneal astigmatism greater 0.75 D undergoing cataract surgery or refractive lens exchange. Preoperative biometry was performed with the IOL-Master (V5.4, Carl Zeiss Meditech, Germany). Subjective refraction, corrected and uncorrected visual acuity, monocular and binocular (near in 30 cm, intermediate in 80 cm and distance [logMar]) as well as binocular defocus curve and rotation stability were evaluated.
Results: So far 44 eyes are enrolled in the study. Three month after surgery median binocular UDVA was 0.0 [logMAR] combined with a median binocular UNVA of 0.0. Even with a near addition of + 4.0 D, median binocular UIVA was 0.2. Only 6 eyes showed a rotation of the tMIOL of a median of 10° and had to be repositioned. After Repositioning no further rotation has been found.
Conclusion: The aberration corrected, pure diffractive, pupil independent and toric multifocal IOL show good refractive results for near, intermediate and distance visual acuity. The rotational stability is very

good. The + 4.0 D near addition in combination with a toric optic offers high spectacle independence and patient satisfaction.

FP-COR-022

Corneal grafts including limbal stem cells in selected patients with stem cell deficiency

Elin Jonsdottir[1,2], Ulf Stenevi[1]

[1]Sahlgrenska University Hospital, Göteborg, Sweden, [2]Södra Älvsborg Hospital, Borås, Sweden

Purpose: The aim of the study was to find out if corneal grafts including limbal stem cells in the periphery is a viable option in treating patients with deficient stem cells, mainly due to severe chemical burns, both in order to maintain a decent visual acuity (VA) and to maintain an eye.
Method: A retrospective design was used. The charts of all patients (in total 19) who had received corneal transplants including limbal stem cells according to Sundmacher's method at the Eye Clinic at Sahlgrenska University Hospital, Gothenburg, between years 1997–2013 were studied. Details such as VA before and after surgery, graft survival and complications were recorded and analysed.
Results: At least 2/3 of the patients had some improvement of VA. Average graft survival lasted 3–5 years. A major complication is glaucoma, which gradually can affect vision adversely. 2 out of 19 patients had to go through evisceration of the eye due to discomfort.
Conclusions: Corneal grafts with limbal stem cells may be a viable option in terms of maintaining severely injured eyes visually functional and pain free.

The degree of limbal stem cell deficiency, as well as the patient's need of visual function and motivation for keeping the eye are factors that will influence the plan of treatment.

FP-COR-040

Innovative soft contact lens for treating Corneal Oedema

Claes Feinbaum[1], Arie Merkovich[2], Ofer Daphnah[2]

[1]Ben Gurion University of the Negev, Beer Sheba, Israel, [2]Kaplan Medical Center, Rehovot, Israel

Purpose: It is a prospective, crossover, randomized study. Primary aim is assessing safety and effectiveness of the Hyper-CL™ lens. Safety will be assessed by evaluating rate of adverse events occurring throughout study period. Effectiveness is assessed measuring changes in corneal thickness.
Setting: Kaplan Medical Center, Rehovot Israel, Barzilai Medical Center, Ashkelon, Israel.
Methods: Twelve eyes with corneal oedema was study population. No treatment occurred 7 days prior to first treatment. Then, each subject was treated, according to a set pattern regimen, 7 days with the Hyper-CL™ lens (treatment A), 7 days with Hyper-CL™ lens + salt solution (treatment B), and 7 days with salt solution only (treatment C). One week break without any treatment between treatments. Clinical evaluation was performed at 7, 14, 21, 28, 35 and 42 days post baseline. Only one subjects' eye will be treated. In case of bilateral corneal oedema the physician identified the most suitable eye for treatment.
Results: No adverse event occurred during the wearing the Hyper-CL™ lens. None of the lens designs caused any side effects rather than normal irritation due to the existence of a contact lens on the cornea. The average baseline pachometry before treatment was 800 μm, after washout with 5% NaCl 722 μm, and after Hyper-CL™ with 5% NaCl 688 μm and again after wash out 666 μm. The result is an average decrease of corneal thickness of 134 μm. Discomfort was noted in 2 eyes and in one eye there was a tendency for the lens to drop out. A slight improvement of vision was noted.
Conclusions: The study encompassed a total of 12 subjects. Safety and efficacy of the Hyper-CL™ have been established. A larger multicentre study is necessary to make sure this method is effective for corneal oedema and corneal dystrophies. The method could be an alternative to other more drastic treatments of corneal disease.

The authors have a financial interest in the product.

FP-COR-083

Terrien's marginal degeneration: Nordic collaborative study

Minna Kurkinen[1], Liv Drolsum[2], Per Fagerholm[3], Jesper Hjortdal[4], Juha Holopainen[1]

[1]Helsinki Eye Lab, University of Helsinki, Helsinki, Finland, [2]Ullevål University Hospital, Oslo, Norway, [3]Department of Clinical and Experimental Medicine-Ophthalmology, Linköping University, Linköping, Sweden, [4]Aarhus University Hospital, Aarhus Universitet, Aarhus, Denmark

Purpose: The objective of our study is to describe the clinical and imaging features of Terrien's marginal degeneration (TMD) and subsequently to define the clinical diagnostic criteria for TMD. Lastly we aim to detect the natural course of this disease.
Methods: Our study design is cross-sectional and includes TMD patients diagnosed in Helsinki University Eye Hospital, Aarhus University Hospital, Sahlgrenska University Hospital and Oslo University Hospital by corneal specialists. Analysis include gender distribution, age, symptoms, as well as clinical, topographical and imaging findings. The analysis is based on descriptive variables due the limited sample size.
Results: We present here clinical findings of Finnish TMD patients. We have identified altogether seven TMD patients; two women and five men. Mean age was 37 years (range 19 to 50 years). All patients were asymptomatic and did not suffer from any systemic autoimmune diseases. Corneal findings were located unilaterally in two patients and bilaterally in five patients. Topographical analysis showed high astigmatism due to severe corneal thinning as evidenced by optical coherence tomography imaging.
Conclusions: TMD is a bilateral corneal disease. Patients are commonly asymptomatic and visit an ophthalmologist because of deteriorated vision due to increasing astigmatism. TMD is thus diagnosed by coincidence. TMD does not seem to be associated with any systemic autoimmune diseases and it is commonly non-progressing. In our series we did not observe any corneal perforations.

FP-COR-084

Penetrating keratoplasty for Gelsolin Type Lattice Corneal Dystrophy (LCD2)

Jaakko Mattila, Tero Kivelä, Kari Krootila, **Juha Holopainen**

University of Helsinki, Helsinki, Finland

Objective: To analyse the prognosis of penetrating keratoplasty for Gelsolin Type Lattice Corneal Dystrophy (LCD2).
Materials/Patients: Twenty-eight patients (35 eyes) with LCD2.
Methods: All patients affected by histopathologically proven LCD2 that received a penetrating corneal transplant (PK) in Helsinki Eye Hospital Cornea service between 1.1.1990 and 1.8.2011 were collected and a retrospective analysis of the patient records was performed. Main outcome measures were best spectacle corrected visual acuity (BCVA), graft survival, incidence of regrafting, intra- and postoperative complications and reason for graft failure.
Results and Conclusion: After a mean follow-up period of 35 $\pm$ 27.9 (range 5–114) months the mean BCVA in LogMAR was 1.06 $\pm$ 0.57 in comparison to the preoperative BCVA of 1.22 $\pm$ 0.4. 21/35 grafts survived to the end of follow-up. Rejection occurred in 7/35 primary grafts (20%) during the follow-up period. Graft failure was due to surface complications in 9 cases and due to surface complications and

rejection in 5 grafts. 7 eyes needed regrafting and 1 eye received three grafts. There were a high number of complications in the early and late postoperative period. There was no clinically significant amyloid deposition in the grafts. In the preoperative assessment the presence of corneal or graft neovascularization was an indicator of high risk of graft failure and poor visual outcome. Age did not affect graft survival. Corneal transplantation can improve vision in a limited number of advanced LCD2 patients. Due to the high failure risk of PK it is important that the physician and the patient have realistic expectations and it might be reasonable to limit corneal transplantation to cases with bilaterally decreased vision.

FP-COR-086

Evaluation of transplanted oral epithelium by confocal microscopy correlates with success rate in aniridia patients

Bogumil Wowra[1], Dariusz Dobrowolski[1], Boguslawa Orzechowska-Wylegala[2], Maria Grolik[1], Ewa Wroblewska-Czajka[1], Edward Wylegala[1]

[1]Department of Clinical Ophthalmology Medical University of Silesia, District Railway Hospital, Katowice, Poland, [2]Department of Cranio-Maxillo-Facial Surgery, Medical University of Silesia, Katowice, Poland

Purpose: To evaluate success of transplanted oral epithelium in corneal surface reconstruction in aniridia patients.

Methods: Twenty-five patients suffered from congenital aniridia accompanied by aniridial keratopathy underwent COMET procedure. Confocal microscopy examination was performed 7, 14 and 28 days after transplantation. Number of cell layers, cell density and stability of epithelium was evaluated.

Results: The survival of transplanted oral epithelium is corelated with early postoperative evaluation of epithelial graft.

Conclusion: Confocal microscopy examination could be useful to predict survival time of oral epithelial graft.

FP-GLA-011

Clinical efficacy and safety of Ex-Press glaucoma device in neovascular glaucoma patients

Rana Hanna, Beatrice Tiosano

Hillel Yaffe Medical Center, Hadera, Israel

Objective: To evaluate the efficacy and safety of Ex-Press implantation in neovascular glaucoma (NVG).

Methods: A retrospective case series that included 4 eyes of 4 patients with NVG who underwent Ex-Press implantation. The device was placed under partial-thickness scleral flap. Changes of intraocular pressure (IOP), the best corrected visualacuity (BCVA), number of anti-glaucoma medications, and postoperative complications were followed up at day 1, month 1, and every 3 months.

Complete success rate defined as reduction in IOP of more than 30% without medications at the last follow up.

Results: The mean follow-up was 10.5 ± 5.7 months (3–15 months). Mean preoperative IOP was 32 ± 5.8 mmHg and mean postoperative IOP at the last visit was 13.0 ± 3.5 mmHg ($p = 0.009$). The control of IOP was achieved at the final follow-up visits in all patients, however, 3 of 4 patients still needed anti glaucoma medications (mean 2 ± 1.4). The BCVA worsened in one eye, and the BCVA was unchanged in the remaining three patients at the final follow -up visits. The postoperative complications included, early phase high IOP and intraoperative hyphema which were all controlled with additional treatment.

Conclusions: The clinical outcome indicates that Ex-Press implantation has a stable IOP lowering effect and low rate of complications, which can be considered as one of the first choices for management of NVG.

FP-GLA-012

Accuracy of optical coherence tomography (OCT) pachymetry in glaucoma patients

Robert Strandås, Marcelo Ayala

Eye Department, Skaraborg Hospital, Skövde, Sweden

Purpose: The aim of the present study was to evaluate the accuracy of the CCT (central corneal thickness) measurements performed by different observers, both with the OCT and the ultrasound pachymetry (USP), in patients suffering from glaucoma. The intention was to determine if OCT might be used as an alternative method for pachymetry instead of today‘s ‘golden standard’ USP.

Methods: Patients who had been previously diagnosed with glaucoma were asked to participate in this prospective study. Glaucoma was defined as patients who had at least 2 repeatable Humphrey visual fields showing glaucoma damage using the software 24-2 and the optic nerve showing typical glaucoma damage. The patients CCT were measured with the OCT (3D OCT-1000; Topcon Corporation) and the USP (Tomey pachymetry, Tomey Corp) by three different examiners. The order of the measurements (OCT/USP or USP/OCT) and which examiner that measured first, second or last, were determined at random.

Results: Seventy-three eyes of 37 patients were included, giving a total of 438 measurements. Average age: 74 yo. The average pachymetry with OCT = 552 ± 37 μm, meanwhile average pachymetry with USP = 533 ± 40 μm. Statistical analysis: Intraclass correlation coefficient; OCT = 0.828 (CI: 0.76–0.88); USP = 0.94 (CI: 0.91–0.96).

Conclusions: Agreement between observers using the OCT or the USP for pachymetry measurements seems to be good. OCT might be used as an alternative method for pachymetry in glaucoma patients.

FP-GLA-045

Ocular surface changes and IOP control using preservative-free Tafluprost

Ingrida Januleviciene, Irmante Derkac

Eye Clinic of Kaunas Medical Academy of Lithuanian University of Health Sciences, Kaunas, Lithuania

Purpose: To evaluate ocular surface changes and IOP control after switching from BAK containing Latanoprost to preservative-free Tafluprost

Design: Observer masked prospective 12 weeks pilot study.

Methods: Thirty open angle glaucoma patients (60 eyes), 26 women (87%) and four men (13%) showing objective ocular surface changes and/or subjective discomfort receiving Benzalkonium chloride (BAK) containing prostaglandine analogue Latanoprost were switched to preservative-free Tafluprost. IOP and ocular surface parameters were measured at Latanoprost baseline and after 2, 6, 12 weeks of Tafluprost treatment.

Results: No statistically significant differences in IOP were observed 2, 6, and 12 weeks after switching to preservative-free tafluprost. Mean IOP at baseline was 16.4 mmHg (SD 2.9), after 2 weeks 16.2 mmHg (2.8), after 6 weeks 16.2 (2.6), and after 12 weeks 16.3 mmHg (2.3). Number of patients expressing discomfort decreased from 100% to 36.6% ($p < 0.05$). Tear osmolarity decreased from 315 to 302 mOsm/l using Tafluprost ($p < 0.05$). BUT increased statistically significantly from 3.7 ± 1.1 seconds to 6.5 ± 1.5 seconds ($p < 0.001$).

Conclusions: Effective IOP control was maintained while subjective discomfort and objective ocular surface changes decreased when patients were switched from BAK preserved Latanoprost to preservative-free Tafluprost.

FP-GLA-046

A novel non-invasive absolute intracranial pressure measurement method using two-depth transcranial Doppler device in glaucoma

Lina Siaudvytyte[1], Ingrida Januleviciene[1], Arminas Ragauskas[2], Laimonas Bartusis[1,2], Alon Harris[1,3]
[1]Eye Clinic of Kaunas Medical Academy of Lithuanian University of Health Sciences, Kaunas, Lithuania, [2]Health Telematics Science Centre of Kaunas University of Technology, Kaunas, Lithuania, [3]Eugene and Marilyn Glick Eye Institute, Indiana University School of Medicine, Indianapolis, IN, USA

Purpose: To present a novel non-invasive absolute intracranial pressure (ICP) measurement method using two-depth transcranial Doppler (TCD) device in glaucoma patients and healthy subjects, which is currently the only available method for absolute ICP value numerical and automatic measurement that does not need an individual patient specific calibration.
Methods: ICP was measured using a novel two-depth TCD device, which is based on simultaneous measurements of blood flow velocities in the intracranial and extracranial segments of the ophthalmic artery (OA). The principle is based on the idea of non-invasive arterial blood pressure measurement, where externally applied pressure to a segment of artery is used to find a balance point when the external pressure is equal to the arterial pressure measure of interest. Pressure balance point is reached when measured blood flow velocity pulsations in both segments of the OA are approximately equal. The absolute ICP measurement value is expressed in mmHg. 50 normal-tension glaucomatous (NTG), 50 high-tension glaucomatous (HTG) and 30 normal eyes were measured with two-depth TCD device (Vittamed 205, Kaunas, Lithuania). The level of significance $p < 0.05$ was considered significant.
Results: ICP was statistically significantly lower in NTG, compared with healthy controls (ANOVA, $p < 0.05$). Patients with NTG also had lower ICP, compared with HTG, however the difference was not statistically significant (ANOVA, $p > 0.05$). Patients with HTG had significantly lower ICP, compared with healthy controls (ANOVA, $p < 0.05$).
Conclusions: The role of the two-depth transcranial Doppler based non-invasive technology for measuring absolute intracranial pressure in glaucoma patients would be innovative and may provide an important aspect currently missing information in glaucoma pathology assessment.

FP-GLA-049

Diode-laser transscleral cyclodestruction in the mode of thermotherapy as the treatment of refractory glaucoma

Ernest Boiko, **Alexey Kulikov**, Vyacheslav Skvortsov, Olga Butikova
Medical Military Academy, Saint-Petersburg, Russia

Purpose: To improve the safety of laser cyclodestruction due to application of the mode of thermotherapy in treating patients with refractory glaucoma.
Methods: Previously, we have proved the safety and efficacy of the laser cyclothermotherapy compared with the traditional cyclophotocoagulation in experimental study in rabbits and in patients with absolute glaucoma. This allowed us to apply this cyclodestruction technique in patients with higher visual functions.

Laser cyclothermotherapy (group I) was performed in 50 patients with refractory glaucoma. Visual acuity was 0.001 in 35 (70%) patients, 0.005–0.1 – 10 (20%) patients and 0.1–0.4 – 5 (10%) patients. 20 diode-laser applications using mode: p = 0.5 W, t = 20 seconds were made. Traditional cyclophotocoagulation (group II) was performed in 20 patients with refractory glaucoma – retrospective data. Visual acuity was 0.001 in 13 (65%) patients, 0.005–0.1 – 5 (25%) patients and 0.1–0.4 – 2 (10%) patients. 20 diode-laser applications were made using mode: p = 1.0–1.5 W, t = 2–5 seconds – in dependence on the ‘popcorn’ effect.
Results: Application of laser cyclothermotherapy compared with the results of applying cyclophotocoagulation in patients with refractory glaucoma made possible to reduce the incidence of expressed postoperative inflammation from 20% to 4% ($p < 0.01$). Number of bleeding complications decreased from 15% to 2% ($p < 0.05$). Visual acuity was reduced only in 6% of the patients in group I and in 35% of the cases in group II ($p < 0.05$). Normalization of IOP persisting for 1 year was achieved in 92% in the cases of application of laser cyclothermotherapy and in 90% – due to the use of cyclophotocoagulation ($p > 0.05$).
Conclusion: Technique of thermotherapy is the most preferable for use in patients with higher visual functions considering the significantly greater safety of laser cyclothermotherapy in comparison with the traditional cyclophotocoagulation.

FP-GLA-057

Structural and functional evaluation of glaucoma patients: A comparative analysis with OCT and HRT

André Vicente, Ana Cabugueira, Arnaldo Santos, Barbara Borges, Vanessa Lemos, Luis Pinto, Joao Feijao, Maria Reina
Centro Hospitalar Lisboa Central, Lisbon, Portugal

Purpose: To compare structural optic disc evaluations performed by optic coherence tomography (OCT) and confocal microscopy (HRT). Structural evaluations were also compared to functional damage using static perimetry (SP).
Methods: The authors present a retrospective study of 90 patients with glaucoma. Structural evaluation was performed with OCT (SPECTRALIS® – Heidelberg Engineering GmbH) and HRT (HRT3® Heidelberg Engineering GmbH) and functional evaluation with the mean defect determined by SP (Octopus Perimetry®, Haag-Streit). Disc area, cup area, rim area, cup/disc area ratio, rim/disc area ratio, cup volume, rim volume, mean cup depth, maximum cup depth and average retinal nerve fibre layer (RNFL) thickness were obtained with HRT. OCT was used to establish average RNFL thickness and thickness by quadrant.
Results: Patients were in average 60, 4 ± 12, 6 years old and 52.2% were men. Average RNFL evaluated by OCT and HRT was positively correlated ($r = 0.476$). This correlation was stronger in the superior temporal area ($r = 0.485$) and weaker in the nasal area ($r = 0.284$). Average RNFL determined by OCT was negatively correlated with cup/disc area ratio ($r = -0.485$), cup area ($r = -0.360$), cup volume ($r = -0.256$) and positively correlated with rim/disc area ratio ($r = 0.560$), rim area ($r = 0.556$), rim volume ($r = 0.546$) evaluated with HRT. MD was also negatively correlated with RNFL thickness. This correlation was stronger when thickness was determined by OCT ($r = -0.643$) and weaker when determined by HRT ($r = -0.385$). All correlations were statistically significant.
Conclusions: Structural damage documented by OCT and HRT is similar and is usually associated with functional damage. Therefore, early detection of optic disc and RNFL structural damage is extremely useful and can be determined by both methods. OCT appears to be more accurate in predicting functional damage.

FP-GLA-063

Intravenous hypertonic saline; the effect on intraocular pressure in exfoliation glaucoma, primary open-angle glaucoma and ocular hypertension patients

Pia Torvinen[1], Mika Harju[1,2], Tero Kivelä[1]
[1]Helsinki University Central Hospital, Helsinki, Finland, [2]Silmäasema, Espoo, Vantaa, Finland

Purpose: To compare the influence of intravenous hypertonic saline (IVHTS) on intraocular pressure (IOP) among patients with exfoliation

glaucoma (ExG), primary open-angle glaucoma (POAG), and ocular hypertension (OHT).
Methods: A prospective, interventional trial. We included patients with IOP 24–30 mmHg. OHT patients lacked topical medication. We excluded patients using oral acetazolamide medication, those with heart or kidney failure, dementia or other systemic condition that markedly decreased physical performance. We injected a bolus of 23.4% IVHTS trough an antecubital vein in a rate 1 ml/s. Dosage was 1.0 mmol/kg sodium. We measured IOP before the bolus (baseline), then every minute after the injection for 10 min, and less frequently for 2 hr.
Results: A total of 44 patients' participated (median age 66 y, range 30–80); 17 patients with ExG, 14 with POAG, and 13 with OHT. The baseline IOP 26.3 (SD, 1.9) mmHg was significantly reduced one minute after treatment to 24.4 (SD, 2.6) mmHg ($p < 0.001$). Maximum IOP reduction was achieved after 10 min, at which time IOP reduction was 8.9 (SD, 2.9) mmHg ($p < 0.001$), and the mean percentage IOP reduction was 33.8% (SD, 11.4%). Mean IOP reduction was similar between groups after 10 min ($p = 0.75$; One-Way ANOVA). Two hours after treatment, the mean IOP reduction was 6.5 (SD, 3.8) mmHg ($p < 0.001$). Within 1–2 min of treatment, 36 patients felt some sensation (pain, pressure) in the infusion arm, and 29 patients reported a feeling of warmth in their head.
Conclusions: Hypertonic saline reduces IOP moderately and rapidly. This reduction seems to be independent of topical medication and glaucoma subgroup, but shows individual variation. IVHTS could be an alternative method to reduce IOP before eye surgery.

FP-NEO-027

Susac syndrome- a case with unusual cardiac, vestibular and imaging manifestations

Rana Hanna, Beatrice Tiosano
Hillel Yaffe Medical Center, Hadera, Israel

Purpose: To report a case of Susac Syndrome (SS) with several unusual features: cardiac involvement with persistent bradycardia, central vestibular involvement with upbeat nystagmus, and the late appearance of MRI white matter changes.
Background: The key clinical features of SS is the triad of encephalopathy, occlusion of the branches of the retinal artery and hearing loss. The cardiac and late MRI findings have not been described.
Methods: THODS- A 38 year-old male was admitted due to severe headache of 2 days duration, dizziness and confusion. The diagnosis of SS was made and he was evaluated by a multidisciplinary team and steroid treatment and IVIG was started.
Results: We describes the development of persistent bradycardia which lasted about 2 weeks, cilioretinal artery occlusion, sensorineural hearing loss and upbeat vertical nystagmus. Repeated MRI showed an ill-defined hyper intense signal of the left hemisphere. A subsequent retinal relapse was documented 2 months after IVIG cessation.
Conclusions: Cardiac involvement is a rare manifestation of SS. Early treatment with combination of steroids and IVIG may be the effective treatment. The presentation with a normal MRI scan could be followed by abnormal MRI scan and subsequent relapse

FP-NEO-053

Amaurosis fugax – characteristics, management and prevalence of significant carotid stenosis

Pia Kvickström[1], Bertil Lindblom[2], Göran Bergström[3], Madeleine Zetterberg[2]
[1]Department of Ophthalmology, Skaraborg Hospital, Skövde, Sweden, [2]Institute of Neuroscience and Physiology, Department of Clinical Neuroscience and Rehabilitation/Ophthalmology, The Sahlgrenska Academy at University of Gothenburg, Mölndal, Sweden, [3]Institute of Medicine, Department of Molecular and Clinical Medicine/Clinical Physiology, The Sahlgrenska Academy at University of Gothenburg, Göteborg, Sweden

Purpose: To describe clinical characteristics, management and prevalence of carotid stenosis in patients with amaurosis fugax (AF).
Method: All patients, $n = 310$, diagnosed with AF and subjected to carotid ultrasound (US) in 2004–2010 at the Sahlgrenska University Hospital, Göteborg, were included. Data were collected from medical records on symptoms, signs, ocular/systemic comorbidity, results of carotid US and carotid endarterectomy (CEA) if any.
Results: The prevalence of significant stenosis (≥ 70%) was 18.5% and 14.6% were subjected to CEA. Significant associations with risk of having ≥ 70% stenosis were age (adjusted Odds Ratio [aOR]: 1.04; 95% Confidence Interval [CI]: 1.01–1.07), male gender (aOR: 2.21; 95% CI: 1.09–4.51), current smoking (aOR: 6.41; 95% CI: 2.71–15.14), hypertension (aOR: 2.21; 95% CI: 1.04–4.67), diabetes (aOR: 3.63; 95% CI: 1.40–9.44) and previous giant cell arteritis (aOR: 10.29; 95% CI: 1.39–76.0). For 76.7% of the patients' visual loss lasted for less than 30 min. Identification of a retinal artery embolus ($n = 1$, 0.4%) was uncommon. A majority of the patients (80.9%) was seen by an ophthalmologist prior to the first US and for 46.3% this was their first medical contact. There was no significant difference in speed of management when comparing specialties initially consulted. Median time to CEA was 21 days and only 43.2% of the patients with significant carotid stenosis had a CEA within 2 weeks.
Conclusion: Most patients diagnosed with amaurosis fugax were seen by an ophthalmologist prior to the carotid US and in a majority of patients this was their first medical contact. Ocular findings were scarce, suggesting that patients may as well have their first consultation with specialists in other fields. Initially consulting an ophthalmologist did not delay the time to US or CEA. The time interval from onset of symptoms to surgery was comparable or shorter than that reported from other countries.

FP-ONC-014

Symptomatic choroidal metastasis as a presenting symptom of lung cancer: A case report

Nina Krøll Angelsen[1], Kristian Fossen[1], Nils Eide[2], Nina Helbekkmo[1]
[1]Universitetssykehuset Nord Norge, Tromsø, Norway, [2]Oslo Universitetssykehus, Oslo, Norway

Introduction: Choroidal metastasis is an unusual presentation of lung cancer, and in this case report we present the effect of initial systemic chemotherapy.
Case Presentation: A 59 year old male presented with visual field loss of his right eye. There was no significant previous medical history. Ocular examination showed best corrected visual acuity of 0.5 Snellen of the right eye and 1.0 of the left eye. Fundus examination revealed a large choroidal mass temporal to the macula. A diagnosis of non-small-cell lung cancer (NSCLC) with cerebral, hepatic, choroidal and lymph node metastasis was made. He was started on systemic chemotherapy with carboplatin and vinorelbine. After the first chemotherapy cycle, ocular evaluation showed regress of the choroidal metastasis. The three following cycles of chemotherapy were delayed and the dose was reduced due to prolonged nadir and an oral infection. Ocular evaluation after the fourth cycle showed progression of both the lung tumour and the choroidal metastasis with vision loss to hand movements only, due to serous retinal detachment with macula off. We chose external beam radiation as secondary treatment modality and the patient is scheduled for this in the near future.
Conclusion: A clinical ocular melanoma was found to be a choroidal metastasis. NSCLC stadium IV has a poor prognosis and the treatment goal is primarily palliative. This case demonstrates a good response to initial chemotherapy with regression of choroidal metastasis, but subsequent progression due to reduced chemotherapy dosage. External beam radiation is also a treatment option and could have been considered earlier in this case.

FP-ONC-023

Tumour regression pattern after brachytherapy for 330 uveal melanomas

Mohammed Mamunur Rashid, Tero Kivelä
Helsinki University Central Hospital, Helsinki, Finland

Purpose: To describe regression patterns of uveal melanoma by thickness and cross-sectional area (CSA).
Methods: A retrospective study of 330 consecutive ciliochoroidal melanomas treated with iodine or ruthenium plaques in 2000–2008 and imaged with the Innovative Imaging-I3-ultrasound. Tumours were measured with Olympus DP-Soft from original digital images. Regression patterns were adapted from Abramson *et al.* (1990).
Results: The regression pattern according to thickness/CSA was D (decrease) in 43%/42%, S (stable, < 15% change) in 5%/2%, I (increase) in 0.3%/0.3%, and M (miscellaneous) in 41%/45% of eyes, respectively. The most common M pattern was DS (decrease-increase) in 16%/16%, followed by Z (zigzag) in 10%/14%, SD (stable-decrease) in 6%/6%, DI (decrease-increase) in 5%/6%, and ID (increase-decrease) in 3%/2% of eyes, respectively. By COMS shape groups (flat-crescent, oval-dome, collarbutton, lobulated), D was the most common pattern (31–63%/33–72% by thickness/CSA). DS was the next most common (22–24%/16–26%), except for oval-dome. Of DI and Z patterns, the former was fairly evenly spread across shapes (5–6%/8–10%) but the latter was observed mainly in flat-crescent and oval-dome groups (15%/17–22%). SD and ID patterns were most common in oval-dome and flat-crescent groups, respectively. Most tumours of S pattern were flat-crescent. Of tumours with an unbroken, indeterminate and broken Bruch's, increasing proportions regressed according to D pattern (39–49–60%/36–57–57% by thickness/CSA). Approximately equal proportions regressed by DS (17–23–18%/15–20–22%) and the other patterns. Tumour thickness regressed notably slower than CSA when the pattern was D. A minor difference to this effect was observed for patterns DI and, especially, DS. No consistent difference was observed for any other pattern
Conclusion: Reduction in tumour thickness and CSA differ especially when regression pattern is D, and this pattern is observed 42–43% of the time. Heterogeneity in regression patterns and tumour shapes must be taken into account when using regression as a statistical variable in outcome analysis.

FP-ONC-025

Management of choroidal melanomas less than 10 mm in largest basal diameter with a 10 mm ruthenium plaque: and update and extended follow-up

Susanna Salkola[1], Jorma Heikkonen[2], Sebastian Eskelin[1], Tero Kivelä[1]
[1]*Department of Ophthalmology, Helsinki University Central Hospital, Helsinki, Finland,* [2]*Department of Oncology, Helsinki University Central Hospital, Helsinki, Finland*

Purpose: To assess local tumour control and mortality after brachytherapy for choroidal melanoma < 10 mm in largest basal diameter (LBD) with the 10 mm ruthenium plaque.
Methods: In 1998–2011, sixty-two choroidal melanomas (all T1aN0M0, stage I) were treated with the 10 mm Ru-CCX plaque. Their median height was 2.1 mm (range, 0.4–5.2) and LBD 7.1 mm (range, 3.3–9.6). The median distance was 3.4 mm (range, 0–10.5) from the optic disk and 2.7 mm (range, 0–10.0) from the foveola. Our standard prescription dose was 100–120 Gy to tumour apex depending on tumour thickness, aiming to deliver a minimum of 250 Gy to sclera.
Results: Seven local recurrences developed a median of 2.3 years (range, 0.6–6.6) after irradiation (median distance from the disk 1.1 mm; range, 0.3–3.75). Five occurred within 3 years from irradiation. Kaplan-Meier recurrence rate was 10% (95% CI, 5–21) by 5 years. Two of these patients developed biopsy-verified liver metastases 7.5 and 44.5 months after diagnosis (0.5 and 12.5 months after recurrence, respectively). A third one later had a solitary brain tumour diagnosed without biopsy verification or evidence of melanoma elsewhere; we nevertheless counted this as metastatic. Kaplan-Meier rate for metastases was 4% (95% CI, 1–14) at 5 years. Nine patients, two with a prior local recurrence and metastases, died a median of 3.7 years (range, 0.3–9.9) after treatment; 5-year all-cause mortality was 10% (95% CI, 4–23) and melanoma-related mortality 4% (95% CI, 1–14).
Conclusions: With 17 added patients and 2 more years of follow-up, the local recurrence rate is slightly higher than in our original report (10% vs 9%) but well within the previously reported confidence interval. Metastatic death rate corresponds to that published for T1a choroidal melanomas in general (6%; Kujala *et al.* 2013).

FP-ONC-072

Frequency and clinical characteristics of familial uveal melanomas in Finland

Tero Kivelä[1], Rosi Wilska[1], Joni Turunen[1], Martin Täll[1], Virpi Raivio[1], Anna-Elina Lehesjoki[2]
[1]*Department of Ophthalmology, Helsinki University Central Hospital, Helsinki, Finland,* [2]*Neuroscience Center, University of Helsinki, Helsinki, Finland*

Purpose: To estimate the frequency and characteristics of familial uveal melanomas in Finland, a high risk region for this cancer.
Methods: Prospective ascertainment of UM patients with a family history between 2008 and 2012 in a national ocular oncology centre managing all UM in Finland. Their clinical characteristics were reviewed and blood was drawn for genetic analysis.
Results: Of 311 new patients with UM, 8 (2.6%; 95% confidence interval, 1.1–5.0) gave a positive family history upon targeted questioning and encouragement to contact relatives. Index cases were 40–83 years old when UM was found (mean, 61) and two of them likely came from the same extended family. The previous cases in the same families were 26–82 years old at the time of diagnosis 3–21 years earlier. The previous patients had the following family relationship to index cases: son, father, mother, grandmother, 2 aunts, uncle, and cousin. Of them, 5 had died of UM and 2 were alive. None of the index patients have so far developed metastases or died. Mutations of BAP1 have been detected in two possibly related families.
Conclusions: Identification and further analysis of familial uveal melanomas will help to understand the emerging genetic background and diversity of uveal melanoma.

FP-ONC-074

Uveal melanoma among Finnish octogenarians

Rana'a Al-Jamal, Tero Kivelä
Helsinki University Central Hospital, Helsinki, Finland

Purpose: To report characteristics of uveal melanoma (UM) among octogenarians.
Materials and Methods: Population-based study of 86 patients 80 years of age or older, treated in 1997–2011 at the Helsinki University Central Hospital for UM. Patients had a minimum of 2 years follow-up.
Results: Of the 86 patients, 36 were female and 50 were male. Median age was 82.9 years, median tumour height was 4.8 mm, median largest diameter was 12.7 mm. 33% were T1, 19% T2, 22% T3, and 26% T4, and 28% were stage I, 32% stage II, 33% stage III, and 6% were stage IV. 70 patients underwent brachytherapy, 12 underwent enucleation, 1 was exenterated and 2 small UM were observed. Median visual acuity was 0.2. At last follow-up, 20 patients had died of metastasis, 21 of other causes, 3 of a 2nd cancer, 26 were alive (median of 5.1 years). The 5-year survival was

75% overall, and 93% for stage I, 79% stage II, and 62% stage III; all patients of stage IV died within 1.3 years. The corresponding all-cause mortalities were 66%, 55%, and 29%.
Conclusion: Taking competing causes of death into account, less than 25% of octogenarians die of UM within 5 years.

FP-ONC-091
Tear protein biomarkers differentiate lacrimal gland tumours from benign conditions

Roger W Beuerman[1,2,3,4], Lay Leng Seah[5], Hannu Uusitalo[4], Lei Zhou[1,2,3,4]
[1]Singapore Eye Research Institute, Singapore, Singapore, [2]Department of Ophthalmology, Yong Loo Lin School of Medicine, National University of Singapore, Singapore, Singapore, [3]SRP Neuroscience and Behavioural Disorder, DUKE-NUS Graduate Medical School, Singapore, Singapore, [4]Proteomics and Personalized Medicine Lab, Ophthalmology, University of Tampere, School of Medicine, Tampere, Finland, [5]Singapore National Eye Centre, Singapore, Singapore

Introduction: Lacrimal gland tumours (LGT) are usually epithelial in origin with approximately 50% classified as benign and 50% as malignant. As the aqueous secretion of the lacrimal gland is the major source of tear proteins and fluid, identification of protein biomarkers in tears is useful for diagnosis of the tumour.
Methods: Tears were collected using Schirmer's strips from a total of 18 patients with different lacrimal gland diseases (malignant LGT: 5 cases; benign LGT: 6 cases; LG inflammation: 7 cases). iTRAQ together with nanoLC-MS/MS was used to quantitatively compare the tear proteomic profiles among three groups as well as the lacrimal gland tissue samples between malignant and benign LG tumour. Gene expression of lacrimal gland tumour tissues was examined using cDNA microarray (Affymetrix Gene 1.0 ST array). ELISA, Immunofluorescent staining and transmission electron microscopy (TEM) were used to confirm the proteomic results.
Results: iTRAQ quantitative proteomics results of tears showed that markedly reduced levels of LG secreted proteins (LCN-1, LYZ, LFT, PIP and PRR4, ect.) and elevated MMP-9 level in tears from malignant LGT compared to two other groups. iTRAQ results of LG tissues also showed that these LG secreted proteins are down-regulated in malignant LGT. TEM showed largely decrease of the number of secretory granules in malignant LGT tissues. ELISA (~100 fold higher in malignant LGT) and immunofluorescent staining results confirmed proteomics results. There is a strong correlation among lacrimal-preferred genes, lacrimal gland secreted proteins in lacrimal gland tissues and tear fluids. They all showed a significant down-regulated expression in lacrimal gland malignant tumour.
Conclusions: The markedly elevated MMP-9 tear level unique to the malignant LGT and could provide a useful diagnostic tool to identify individuals with lacrimal gland malignant tumours and could easily be implemented as a clinical decision making device.

FP-PAE-008
Awareness and attitudes among parents toward eye diseases in their children with Down syndrome: A cross-sectional survey in Jeddah

Firas Madani, Nizar Alhibshi, Mehenaz Hanbazazh
Faculty of Medicine, King Abdulaziz University, Jeddah, Makkah, Saudi Arabia

Purpose: Ocular problems affecting children with Down syndrome (DS) tend to be more frequent and severe than in other children. This paper aimed to investigate the awareness and attitudes of parents of children with DS toward eye diseases that affect their children.
Methods: A cross-sectional survey was conducted on the parents of children with DS who attended the World Down Syndrome Day event in Jeddah, Saudi Arabia. Questionnaires were used to assess parents' knowledge and attitudes toward common eye diseases affecting children with DS. Data were analysed using the Statistical Package for the Social Sciences.
Results: A total of 72 parents were surveyed. Only 49 parents (74.2%) reported ever taking their child for a visit to an ophthalmologist, and only 27 of the parents (55.1%) reported taking their child for a follow up visit. The most common sources of information reported by the parents were the Internet (n = 41; 56.9), ophthalmology clinics (n = 28; 38.9%), and books (n = 22; 30.6%). Many parents did not perceive common ophthalmologic disorders in their children with DS; the proportion of positive responses to questions that identified common eye conditions was 48.5%.
Conclusion: Eye care of children with DS is insufficient, mainly owing to parents' poor perception and lack of awareness about ocular disorders that commonly occur in children with DS, thus justifying the need to educate parents of children with DS in order to promote early diagnosis and proper follow up of cases with ocular disorders.

FP-PAE-029
Morning Glory Disc Anomaly in children and adolescents. Prevalence, ocular characteristics and associated somatic and behavioural disorders

Dylan Ceynowa[1], Monica Olsson[2,3], Ulla Ek[4], Urban Eriksson[3], Ronny Wickström[5], **Kristina Teär Fahnehjelm**[2,6]
[1]Karolinska Institutet, Stockholm, Sweden, [2]Department of Clinical Neuroscience, Karolinska Institutet, Stockholm, Sweden, [3]St. Erik Eye Hospital, Stockholm, Sweden, [4]Department of Special Education, Stockholm University, Stockholm, Sweden, [5]Neuropediatric Unit, Astrid Lindgren Childrens Hospital, Department of Womens and Childrens Health, Stockholm, Sweden, [6]At the time of the study, Department of Pediatric Ophthalmology St. Erik Eye Hospital; and now Department of Ophthalmology Vrinnevi Hospital, Norrköping, Sweden

Purpose: Morning Glory Disc Anomaly (MGDA) is a congenital vision-threatening optic disc anomaly. The purpose of this study was to report the prevalence of MGDA in children and adolescents, and to describe ocular characteristics and associated neurological, behavioural, and/or somatic disorders in patients with MGDA.
Methods: This is a population-based cross sectional study. Patients with MGDA were identified from the data base at the department of paediatric ophthalmology and were invited to take part of the study. Clinical ophthalmological assessments including best corrected visual acuity, ocular alignment, refraction in cycloplegia, ophthalmoscopy, fundus photography, optical coherence tomography (OCT) and ocular motor score (OMS) were performed and evaluated. Neurological examinations and behavioural/developmental screening were performed and analysed.
Results: Eleven patients aged 2 to 19 years were detected and included. The prevalence of MGDA was calculated to be 2.4/100 000. All cases were unilateral. The best-corrected visual acuity in MGDA eyes ranged from hand motion to 0.65. The OCT examinations demonstrated per papillary atrophy and oedema. Three patients had coexisting capillary haemangioma in the orbit or neck regions. One patient of these had PHACES, a neurocutaneous syndrome, and malformation of the internal carotid artery. Another patient had corpus callosum agenesis and cerebrovascular anomalies. Attention deficit disorder and neurological dysfunction was identified in one patient. Behavioural/developmental screening was normal in all patients.
Conclusions: MGDA is a rare, congenital disc anomaly. Severe somatic comorbidities may not be discovered at ophthalmological examination at the time of diagnosis. This calls for attention from the paediatric ophthalmologist and referral for further neuropaediatric evaluation.

FP-PAE-030

Visual functions and ocular characteristics in children with cochlear implants due to congenital cytomegalovirus infection

Kristina Teär Fahnehjelm[1,2], Monica Olsson[1,3], Eva Karltorp[4], Cecilia Fahnehjelm[5], Ilona Lewensohn Fuchs[6]

[1]Department of Clinical Neuroscience, Karolinska Institutet; At the time of the study Department of Paediatric Ophthalmology, St.Erik Eye Hospital, Stockholm, Sweden, [2]Department of Ophthalmology, Vrinnevi Hospital, Norrköping, Sweden, [3]Department of Pediatric Ophthalmology, St.Erik Eye Hospital, Stockholm, Sweden, [4]Department of Clintec, Karolinska Institutet and Department of Otorhinolaryngology, Karolinska University Hospital, Stockholm, Sweden, [5]Karolinska Institutet, Stockholm, Sweden, [6]Division of Clinical Microbiology, Department of Laboratory Medicine, Karolinska Institutet and Department of Clinical Microbiology, Karolinska University Hospital, Stockholm, Sweden

Purpose: To study visual function and ocular characteristics in children with cochlear implants due to severe hearing impairment caused by congenital cytomegalovirus (cCMV) infection.

Methods: Clinical ophthalmological assessments included best corrected visual acuity, ocular alignment, Ocular Motor Score (OMS), bio microscopy, refraction and fundus photography. Children with connexin 26 deficiencies (Cx26), a genetic cause of hearing impairment, were used as controls.

Results: Twenty-six children with cCMV were included. Their median age was 8.3 years (range 1.4–16.7). Thirteen cx26 controls had a median age of 5.6 years (range 1.7–12.5). Five cCMV children of 26 (19%) had unilateral chorioretinal macular scars. None of the children with Cx26 deficiency had chorioretinal scars ($p = 0.15$). Ocular motility problems were more common among children with cCMV but the difference was not significant ($p = 0.19$). The vestibuloocular reflex (VOR), was more often pathological in children with cCMV ($p < 0.05$).

Conclusion: Central chorioretinal scars and ocular motility disturbances were common in children treated with cochlear implants due to severe hearing impairment caused by cCMV infection. Ophthalmological assessments including extended ocular motor assessment and VOR are advised in these children.

FP-PAE-082

Macular function measured with mfERG in prematurely-born children at school-age

Hanna Åkerblom[1], Sten Andreasson[2], Gerd Holmström[1]

[1]Department of Neuroscience, Ophthalmology, Uppsala University, Uppsala, Sweden, [2]Department of Ophthalmology, Lund University, Lund, Sweden

Purpose: Children born preterm have affected visual functions compared to children born at term. We have previously shown morphological changes, measured with optical coherent tomography (OCT), of prematurely-born children. The aim of this study is to evaluate the function of the macula with multifocal electroretinogram (mfERG) and investigate correlations between macular function and visual acuity (VA), gestational age (GA), birth weight (BW) and central macular thickness measured with OCT.

Material and Methods: A preterm group of ten children, 9–15 years old, born before 32 weeks of gestation, were evaluated with mfERG (VERIS, EDI) and OCT (CIRRUS, Carl Zeiss, Meditec). Their mean GA was 29 weeks and mean BW was 1273 g. Four children had no retinopathy of prematurity (ROP) and six had mild ROP in the neonatal period. A group of 11 full-term children, 8–19 years old, with normal VA, were used as controls. A program with 103 hexagone stimulation was used and the amplitudes and implicit times of the first positive peak, P1, were presented in five concentric rings and a 'sum of all groups'.

Results: The amplitudes of the P1 response of ring 1–5 and 'sum of all groups' were significantly reduced in the preterm group compared to controls. In these preliminary results there were no correlation between P1 amplitude and implicit times with VA, GA, BW or central macular thickness.

Conclusions: The central macular function, measured with mfERG, is reduced in school-aged, prematurely born children, compared to children born at term. These results could be one reason why children born preterm have affected visual functions even if they had no or only mild ROP in the neonatal period. A possible explanation is that preterm birth *per se* affects the development of the macula leading to permanent morphological as well as functional changes.

FP-RET-021

Autophagy in the pathogenesis of age-related macular degeneration

Kai Kaarniranta[1,2]

[1]University of Eastern Finland, Kuopio, Finland, [2]Kuopio University Hospital, Kuopio, Finland

Purpose: The pathogenesis of age-related macular degeneration (AMD) essentially involves chronic oxidative stress, increased accumulation of lipofuscin in retinal pigment epithelial (RPE) cells and extracellular drusen formation, as well as the presence of chronic inflammation. The presence of lipofuscin decreases lysosomal enzyme activity and impairs autophagic clearance of damaged proteins which should be removed from cells. Proteasomes are another crucial proteolytic machine which degrade especially cellular soluble proteins damaged by oxidative stress. Molecular mechanisms behind protein aggregations are weakly understood. There is intriguing evidence suggesting that protein SQSTM1/p62, together with autophagy, has a role in the pathology of different degenerative diseases. It appears that SQSTM1/p62 is a connecting link between autophagy and proteasome mediated proteolysis, and expressed strongly under the exposure to various oxidative stimuli.

Methods: The SQSTM1/p62, LC3 and ubiquitin protein levels and localization were analysed by western blotting and immunofluorescense in ARPE-19 cells. Confocal and transmission electron microscopy were used to detect cellular organelles and to evaluate the morphological changes. Cell viability was measured by colorimetric and FACS assays. AMD human donor samples were stained for SQSTM1 and ubiquitin.

Results: Under proteasomal inhibition SQSTM1/p62 was up-regulated in RPE cell culture condition. Proteasomal inhibition caused accumulation of SQSTM1/p62 which bound irreversibly to perinuclear protein aggregates. The addition of the AMPK activator AICAR was pro-survival and promoted cleansing by autophagy of the former complex suggesting that SQSTM1/p62 is decreased through autophagy-mediated degradation. Interestingly, when compared to human controls, AMD donor samples showed strong SQSTM1/p62 accumulation in the drusen rich macular area.

Conclusions: Autophagy dysregulation leads to retinal pigment epithelium dysfunction and development of age-related macular degeneration. The autophagy clearance system might represent a potential future therapeutic target for AMD.

FP-RET-024

Hyperreflective dots detected in the retina using spectral domain optical coherence tomography

Simona Stech[1,2], Kristina Kuoliene[1,2], Andrius Cimbalas[1,2], Rimvydas Asoklis[1,2]

[1]Vilnius University Faculty of Medicine, Vilnius, Lithuania, [2]Centre of Eye Diseases, Vilnius University Hospital Santariskiu Klinikos, Vilnius, Lithuania

Purpose: To investigate morphologic changes in posterior segment diseases using spectral-domain optical coherence tomography (SD-OCT). To discuss the role and characteristics of hyperreflective dots (HRD) in retinal inflammatory diseases and choroidal neovascularization (CNV).
Methods: Patients with posterior segment diseases with or without choroidal neovascularization (GNV) were retrospectively reviewed. Thorough ophthalmologic examination and SD-OCT were performed during the first visit and follow-up. HRD behaviour was evaluated in the retina with SD-OCT pre- and post-treatment.
Results: In all cases, various amounts of HRD were detected in the foveal or parafoveal area, and in one case near the optic disc prior to treatment. HRD accumulated densely especially surrounding the area of fluid accumulation. During follow-up, progressive resolution of the hyperreflective dots was observed on SD-OCT. HRD reduction was observed mainly in cases of complete macular oedema resolution and correlated greatly with visual acuity improvement.
Conclusions: HRD seen in retinal inflammatory diseases on SD-OCT behave similarly as in wet AMD and diabetic maculopathy. Association between VA improvement and decrease in the amounts of HRD could be used as a helpful treatment indicator in both chorioretinitis and choroidal neovascularization.

FP-RET-032

Macular thickness measurements in normal and diabetic eyes before and after meal

Khalil Ghasemi Falavarjani, **Joobin Khadamy**, Mehdi Modarres, Hamid Safi, Foad Amirkourjani
Eye Research Center, Rassoul Akram Hospital, Iran University of Medical Sciences, Tehran, Iran

Purpose: To evaluate the retinal thickness changes in normal eyes and in diabetic macular oedema after meal.
Methods: In this prospective nonrandomized study, retinal thickness of diabetic patients with clinically significant macular oedema (CSME) and normal subjects were measured after 7 hr of fasting and repeated 2 hr after breakfast.
Results: Thirty six eyes of 20 diabetic patients and 38 eyes of 19 normal subjects were evaluated. In normal subjects, the mean central subfield thickness (CST) and maximum retinal thickness (MRT) measurements were statistically similar before and after meal (both $p = 0.9$). In diabetic eyes, the mean CST and MRT significantly decreased after meal (mean change of -10.3 ± 14.3 μm and -13.1 ± 12.7 μm, respectively, $p < 0.001$). A decrease in CST and MRT values was found in 23 (63.8%) and 28 (77.7%) eyes, respectively, and no eye had an increase in retinal thickness measurements. Significant correlation was found between CST and MRT change and fasting thickness measurements ($p = 0.001$ and $p = 0.01$, respectively) and intraretinal cystic spaces ($p = 0.001$ and $p = 0.03$, respectively). Mean MRT change was significantly higher in the presence of subretinal fluid ($p = 0.01$).
Conclusion: Retinal thickness changes after meal should be considered in the management of the patients with CSME.

FP-RET-056

Duration of lactation is associated with retinal vascular calibres: the Tromsø Eye Study

Therese von Hanno[1,2], Geir Bertelsen[3,4], Ellisiv B Mathiesen[1]
[1]Brain and Circulation Research Group, Department of Clinical Medicine, Faculty of Health Sciences, UiT The Arctic University of Norway, Tromsø, Norway, [2]Department of Ophthalmology, Nordland Hospital, Bodø, Norway, [3]Research Group of Epidemiology of Chronic Diseases, Department of Community Medicine, Faculty of Health Sciences, UiT The Arctic University of Norway, Tromsø, Norway, [4]Department of Ophthalmology and Neurosurgery, University Hospital of North Norway, Tromsø, Norway

Purpose: Lactation may have beneficial effects on maternal cardiovascular risk factors, disease and mortality. We wanted to investigate if duration of lactation was associated with changes in retinal vascular calibres.
Methods: Population-based cross-sectional study including 3145 women aged 38–87 years who participated in the Tromsø Eye Study, 2007–2008. Retinal arteriolar calibre (central retinal artery equivalent) and retinal venular calibre (central retinal vein equivalent) were measured computer-assisted on retinal photographs. Data on parity and duration of lactation was collected from questionnaires.
Results: Among parous women younger than 60 years, mean duration of lactation per child was inversely associated with retinal venular calibre (p-trend, 0.003). There was a dose-dependent relationship – when compared to women with mean duration of lactation < 3 months, women with duration 3–6 months had 3.8 (95% CI 7.6, 0.1) μm thinner venules and women with duration > 6 months had 5.7 (95% CI 9.4, 2.0) μm thinner venules. Retinal arteriolar calibre was 2.7 (95% CI 5.1, 0.2) μm thinner in women with duration of lactation ≧ 3 months compared to duration < 3 months, while we could not demonstrated a dose-dependent relationship. These differences remained when adjusted for established cardiovascular risk factors. Among parous women ≧ 60 years duration of lactation was not associated with retinal vascular calibres. Nulliparous women had no significant different retinal vascular calibres compared to parous women.
Conclusions: Longer duration of lactation was associated with thinner retinal venular and arteriolar calibre in parous women younger than 60 years.

FP-RET-068

Initial improvement after converting to afliberecept is substantially diminished when increasing control intervals from 4 to 8 weeks

Øystein Kalsnes Jørstad[1], Geir Aksel Qvale[1], Morten C Moe[1,2]
[1]Department of Ophthalmology, Oslo University Hospital, Oslo, Norway, [2]University of Oslo, Oslo, Norway

Purpose: To examine the structural and functional response after converting to afliberecept (Eylea) according to label dosing in patients with exudative age-related macular degeneration (wet AMD) and persistent macular oedema on Avastin or Lucentis therapy every fourth week.
Methods: Patients converted to Eylea were given 3 monthly injections (loading dose) before extending to 8 weeks control interval according to label dosing. LogMar visual acuity, SD-OCT (RS-3000 Nidek Inc.) and clinical examination were recorded in 50 consecutive patients.
Results: After the loading dose there was a significant improvement of mean central retinal thickness ($p < 0.05$). Forty percent had a dry macula on OCT and another 30% showed structural improvement of retinal oedema. Thirty percent of patients with pigment epithelium detachment (PED) at baseline also had a decreasing PED. There was a non-significant trend towards improvement in LogMAR visual acuity. When the injection interval was extended to 8 weeks most of the improvement seen during the loading phase was lost.
Conclusions: Converting to Eylea according to the label dosing was associated with a structural improvement during the loading phase of the treatment. When the injection interval was extended to 8 weeks most of this improvement was lost. The non-inferiority results on q8 dosing from the VIEW-studies for naive wet AMD does not necessarily reflect non-inferiority for all wet AMD subgroups.

FP-RET-081

A prospective pilot study of the effects of panretinal photocoagulation delivered with a multi-spot laser on retinal sensitivity and driving eligibility in patients with diabetic retinopathy

Michel Michaelides[1,2], Mala Subash[1], Ahmed Samy[1], Serafeim Antonakis[1], Dania Qatarneh[1], Tunde Peto[1,2], Louisa Wickham[1]
[1]Moorfields Eye Hospital, London, UK, [2]UCL Institute of Ophthalmology, London, UK

Purpose: The gold standard treatment for severe non-proliferative and proliferative diabetic retinopathy (DR) is panretinal photocoagulation (PRP) that may have a damaging effect on retinal function. Multi-spot laser systems are increasingly used and believed to cause less collateral retinal damage. This study describes the baseline characteristics of the cohort of recruited patients who have undergone Valon-multispot laser and have had detailed retinal imaging and retinal sensitivity testing pre- and post-treatment.
Methods: Laser-naïve patients requiring bilateral PRP were enrolled. All subjects underwent standard clinical examination and SD-OCT. Ultra-widefield fundus photography and fundus fluorescein angiography using the Optos system were performed at baseline and at 6 months from initial visit. Moorfields Reading Centre is grading the ETDRS severity of DR, severity of peripheral capillary closure, and laser burn coverage.

Retinal sensitivity was assessed using Binocular Estermann Driving Visual Fields on the Humphrey VFA and full-field static testing on the Octopus 900. The Valon (Dual Laser Ltd, Finland) multi-spot laser was used to deliver standardised PRP.
Results: Forty-seven patients have been recruited to date (total = 50), of these 20 were males, 16 had Type 1 diabetes and the average age was 47.3 with the average duration of diabetes being 17.6 years. Of the 47 recruited, 25 have completed 6-months follow-up to date. The driving visual fields remained unchanged in 24. Provisional results suggest a significant likelihood of maintaining the visual field criteria to hold a UK driving license following bilateral multi-spot PRP.
Conclusion: Multi-spot PRP lasers are being used increasingly worldwide. Interim data from this study suggests that a significant proportion of patients will retain driving eligibility following PRP. Further analysis comparing Optos-image derived retinal ischaemia area and retinal laser coverage, with retinal sensitivity data will be valuable and add to our understanding of peripheral retinal sensitivity pre- and post-laser therapy.

FP-VIT-020

Endophthalmitis after small-gauge vitrectomy: A retrospective case series

Marcin Czajka[1], Eva Byhr[2], Göran Olivestedt[3], Eva Olofsson[4]
[1]Department of Ophthalmology, University Hospital, Linköping, Sweden, [2]Department of Ophthalmology, Sahlgrens University Hospital, Gothenburg, Sweden, [3]Department of Vitreoretinal Diseases, St. Eriks Eye Hospital, Stockholm, Sweden, [4]Department of Ophthalmology, Umeå University Hospital, Umeå, Sweden

Purpose: The objective of this study was to present anatomical and functional results of the treatment in patients with endophthalmitis after small-gauge vitrectomy.
Methods: Retrospective, multicentre case series involving patients with endophthalmitis in Sweden between 2008 and 2012.
Results: A total 23 patients (23 eyes). Indications for small-gauge vitrectomy included epiretinal membrane (n = 13), retinal detachment (n = 5), others (n = 5). Surgical technique included 23G and 25G vitrectomy (22:1), mean surgical time (42.5 min ± 22). Postoperative hypothony < 7 mmHg (n = 2). Days to endophtahlmitis presentation varied between 1 and 21 (mean 6 ± 6). Presenting BCVA were hand motion (n = 13), light perception (n = 7), counting fingers (n = 2), no data (n = 1). Treatment methods included: tap injection (n = 6), tap injection and vitrectomy (n = 2), vitrectomy with antibiotics (n = 15). Culture showed Coagulase-negative staphylococcus (n = 9), Staphylococcus aureus (n = 3), β-haemolytic Streptococcus group C (n = 1), Bacillus Cereus (n = 1), Enterococcus (n = 1), negative (n = 8). Anatomical results included evisceration (n = 1), phthisis (n = 1), globe intact (n = 21). Mean BCVA at last follow-up were 0.5, 83% (19 of 23) of eyes achieved a visual acuity ≥ 0.1 and 52% (12 of 23) had final visual acuity ≥ 0.5
Conclusions: In spite of possible devastating complications of endophthalmitis after small-gauge vitrectomy our study showed good anatomical and functional results with no light perception and evisceration in only 7% (2 of 23) of the eyes.

FP-VIT-050

Phacovitrectomy for open globe injures with intraocular foreign bodies and traumatic cataracts

Ernest Boiko, Sergey Churashov, **Alexey Kulikov**, Natalia Haritonova
Medical Military Academy, Saint-Petersburg, Russia

Purpose: To evaluate the visual outcomes of phacovitrectomy for patients with penetrating open globe injures (OGI) with intraocular foreign bodies (IOFBs) and traumatic cataracts.
Methods: This was a prospective, nonrandomized, observational sequential case-series study on 15 patients (15 eyes) who had undergone phacovitrectomy for penetrating OGI with IOFBs (grade C, zones I, II) and traumatic cataracts during the 2013 year at the Department of Ophthalmology, Military Medical Academy. Combined surgical procedure included phacoemulsification of traumatic cataract, pars plana 25 G vitrectomy, posterior hyaloid membrane peeling, consecutive IOFB removal through corneal tunnel and intraocular lens implantation. The follow up period was 6–12 months. Preoperative and postoperative data included best corrected visual acuity (BCVA), slit-lamp bio microscopy.
Results: The preoperative mean BCVA was 0.03 ± 0.02 in dependence on injury size and location, lenticular opacity degree, presence of vitreous haemorrhage and macular damage. Postoperative mean BCVA significantly improved and was 0.57 ± 0.11 in 12 patients, having no macular damage and postoperative proliferative vitreoretinopathy. In two patients postoperative mean BCVA was 0.35 ± 0.05 where IOFB was impacted into the paramacular retina and formed the rough chorioretinal scars. One patient had corneoscleral (zone II) OGI, that after phacosurgery was followed by proliferative vitreoretinopathy and retinal detachment. Consequently one more reattachment surgical procedure was applied and 6 months postoperatively BCVA was 0.1.
Conclusion: Combined vitrectomy with phacoemulsification in penetrating OGI with IOFBs and traumatic cataracts produced short-term favorable visual outcomes. The main adverse factor of visual recovery in corneoscleral OGI is proliferative vitreoretinopathy, that is why the patients with OGI of zone II need more careful and long-term postoperative observation.

FP-VIT-051

Susac's syndrome detected by ultra-widefield fluorescein angiography

Panagiotis Salvanos[1,2], Morten C Moe[1,2], Nils Eide[1], Andreas Wigers[1], Ragnheidur Bragadottir[1,2], Emilia Kerty[2,3]
[1]Department of Ophthalmology, Oslo University Hospital, Oslo, Norway, [2]University of Oslo, Oslo, Norway, [3]Department of Neurology, Oslo University Hospital, Oslo, Norway

Aim: To present a case of Susac's syndrome detected by ultra-widefield fluorescein angiography (UW-FA) and autofluorescence.

Methods: A 34-year-old myopic woman was referred to the Department of Ophthalmology, Oslo University Hospital due to rhegmatogenous retinal detachment in her left eye (OS). Preoperatively, it was noticed that the retina was avascular in 2/3 of the periphery in both eyes (OU) and confrontation visual fields showed constricted outer borders.

The patient was operated with scleral buckling with the use of encircling band. Laboratory work-up for vasculitis was obtained.

During 1-year follow-up, the ischemic retina progressed further towards the posterior pole and the patient developed secondary neovascularization on the iris, anterior chamber angle and optic disc OU. During the same period, she observed significant hearing loss in both ears and admitted of having low-grade depression.

UW-FA was obtained both preoperatively and during follow-up.

Results: Vasculitis work-up was negative, including temporal artery biopsy, carotid artery Doppler examination and brain MRI. UW-FA showed multiple branch retinal artery occlusions (BRAOs) with Gass plaques, leakage was observed during late phase and obvious progression during follow-up. The co-existence of multiple progressing BRAOs with hearing loss and organic psychosyndrome led to the diagnosis of Susac's syndrome. The retina remained attached during follow-up with visual acuity of 1.0. The patient was treated with high dose intravenous and oral steroids, anticoagulants and panretinal photocoagulation. After treatment, both hearing function and psychological status returned to normal and the proliferative disease regressed.

Conclusion: UW-FA shows the multiple BRAOs and the avascular peripheral retina, and is highly assisting in documenting and following-up patients with peripheral retinal pathology. Suspicion of Susac's syndrome should be raised in middle-aged women with multiple BRAOs, hearing loss and neurological/psychological problems.

PP-CAT-004

Presbidrops™ (pp) – a medical treatment to correct Presbyopia

Claes Feinbaum

Ben Gurion University of the Negev, Ashkelon, Israel

Presbidrops™ is a medical treatment to correct presbyopia by means of ophthalmic drops. A study has been undertaken in order to justify further research of the actions of this eye-drop and if it really helps in overcoming the problems of intermediate and near vision.

The pharmacological treatment has 2 purposes namely a stimulation of the parasympathetic innervation (accommodation) and prolonging the effect of the parasympathomimetic action (NSAID).

Resulting in:

1 Increased Depth of Focus
2 A decrease of the Higher Order Aberrations

Patient is between 40 and 80 years of age without any ocular pathology except phorias and glaucoma and intra-ocular lenses. Distance refraction should be between + 0.75 to −0.75 D sphere and astigmatism less than −2.00. Dosage and periodicity is decided individually.

Number of eyes participating in study was 81.

Mean pupil diameter 4.1 mm and after instillation 2.7 mm, standard deviation 1.110 and p-value 0.8150. Mean UCVA before 0.8 and after instillation 1.0, standard deviation 0.1907 and p-value 0.8361. Mean NUCVA 0.54 and after instillation 0.8 standard deviation 0.1907 and p-value 0.8361. DOF 1.6 D and after 2.8 D, standard deviation 0.8288 and p-value 0.3760. The safety is excellent, all eyes improved both for distance and near.

Adverse reactions: 1 patient nausea quickly relieved, 1 patient dryness gradually disappearing and 1 patient burning feeling gradually disappearing.

Thus a topical treatment to correct presbyopia is a safe and easy way to eliminate the use of glasses in presbyopic patients. It is completely reversible. A complete and careful examination of each patient is necessary and you have to know that it is not suitable for all presbyopic patients.

PP-CAT-034

Clinical results with a 'Blended Vision' approach to ReSTOR® IOL implantation

Kjell Gunnar Gundersen

Privatsykehuset Haugesund, Haugesund, Norway

Purpose: To clinically evaluate a 'Blended Vision' approach to AcrySof® IQ ReSTOR® implantation, using a + 2.5 D lens in one eye and a + 3.0 D lens in the other. In addition, to compare the performance of this 'Blended Vision' option to available data from bilateral ReSTOR + 2.5 and bilateral ReSTOR + 3.0 patients who were previously treated at the same site.

Methods: Post-operative visual acuity and quality of vision data were recorded from patients who had previous implantation of an AcrySof® IQ ReSTOR® + 2.5 D lens in one eye and + 3.0 D lens in the other. Uncorrected and best distance-corrected visual acuity at distance (4 m), intermediate (60 cm) and near (40 cm), as well as low contrast acuity at 4 m, were measured. Each patient's preferred reading distance was recorded, along with the visual acuity at that distance. Quality of vision metrics were also collected. Data from these patients were compared to available data from a previous study that included patients implanted bilaterally with either the ReSTOR + 2.5 D or ReSTOR + 3.0 D lenses.

Results: Data from a total of 21 Blended Vision patients were collected. Refractive outcomes showed excellent predictability with 93% of eyes within 0.5 D of the target refraction. Uncorrected binocular distance visual acuity was better than 0.1 logMAR (0.8 decimal acuity) in 95% of patients. Best distance-corrected VA was equivalent to the binocular + 2.5 patients at intermediate and equivalent to the binocular + 3.0 patients at near. Binocular distance low contrast acuity was similar to that of the binocular + 2.5 patients. Reading distance was similar to the binocular + 3.0 patients and all Blended Vision patients had 0.2 logMAR (0.67 decimal) or better VA at their preferred reading distance.

Conclusions: Blended Vision with the ReSTOR + 2.5/+ 3.0 appears to provide patients with the advantages of both lenses, rather than a compromise between the binocular + 2.5 and binocular + 3.0 options.

PP-CAT-035

Diffractive multifocal IOL's. A comparative study of Finevision vs ReSTOR 2.5 & 3.0D

Kjell Gunnar Gundersen

Privatsykehuset Haugesund, Haugesund, Norway

Purpose: To clinically evaluate Finevision Trifocal MFIOL to AcrySof® IQ ReSTOR® 2.5 and 3.0 D implantation.

Methods: Post-operative visual acuity and quality of vision data were recorded from patients who had previous bilateral implantation of AcrySof® IQ ReSTOR® + 2.5 D, AcrySof® IQ ReSTOR® + 3.0 D or Finevision Trifocal lenses. Uncorrected and best distance-corrected visual acuity at distance, intermediate and near, as well as low contrast acuity at 4 m, was measured. AcrySof® IQ monofocal lenses were used as control IOL's in the study.

Each patient's preferred reading distance was recorded, along with the visual acuity at that distance. Quality of vision metrics were also collected.

Results: Data from 96 patients were collected, 32 from each group. Refractive outcomes showed excellent predictability for both ReSTOR and Finevision eyes. Uncorrected binocular distance visual acuity was better than 0.1 logMAR (0.8 decimal acuity) in 95% of patients, with no statistically difference between groups. Finevision eyes showed the best distance-corrected VA at both intermediate

(60 cm) and at near (40 cm) compared to AcrySof® IQ ReSTOR® + 2.5 D and AcrySof® IQ ReSTOR® + 3.0 D near. Binocular distance low contrast acuity was similar between Finevision and AcrySof® IQ ReSTOR® + 2.5 D patients, whereas AcrySof® IQ ReSTOR® + 3.0 D eyes performed slightly worse. Preferred reading distance was similar between Finevision and AcrySof® IQ ReSTOR® + 3.0 D eyes (approx. 42 cm), compared to approx. 50 cm in AcrySof® IQ ReSTOR® + 2.5 D eyes. Visual acuity at preferred reading distance showed no statistically difference between groups. Quality of vision was significantly better for Finevision eyes compared to AcrySof® IQ ReSTOR® + 3.0 D eyes but not significantly better than AcrySof® IQ ReSTOR® + 2.5 D eyes.
Conclusions: Finevision eyes compared favorably to both AcrySof® IQ ReSTOR® + 2.5 D and AcrySof® IQ ReSTOR® +3.0 D at both intermediate and near reading tests. Finevision eyes showed better Quality of vision compared to AcrySof® IQ ReSTOR® +3.0 D eyes.

PP-CAT-036

The limitations of computer displays in visual acuity measurement

Ville Voipio, Reetta Forsvik, **Heikki Forsvik**
Tampere University of Techology, SDepartment of Signal Processing, Tampere, Finland

Purpose: The Nordic countries have a long history in visual correction and visual acuity measurement. The current trend in visual acuity testing is towards the use of computerized systems. Computerized visual testing offers several possibilities in improving availability and reliability, and computer displays are ubiquitous in health care. Relatively little attention has, however, been paid to the technical requirements for the displays. The purpose of this study is to find out the technical limitations of modern computer displays when used in visual acuity measurement.
Methods: The requirements for a computer display used in visual acuity testing are established mathematically by modelling the optical chain from the visual acuity target to patient's eye. The pixellated nature of computer displays is taken into account.

The actual performance of modern computer displays is determined by using their published specifications. A few typical displays and different optotypes are analysed to find the acceptable parameters.
Results: The resolution of a typical computer display is inferior to that of a paper chart. If the angular separation of the pixels is less than approximately 1/8 arc min, the display can be used up to 2.0 visual acuities without display-related errors. At 5 m viewing distance this translates into 0.2 mm pixel pitch, which can be obtained with a high-resolution display. Large screens or low-resolution displays have lower resolutions.

The resolution limits become more difficult to meet if a computer display is used to determine the near visual acuity at a close distance. With a typical computer monitor at 40 cm distance the maximum testable visual acuity is in the range of 0.2. . . 0.4, which is below normal vision at that distance.
Conclusions: Standard computer displays can be used in visual acuity testing in many settings, but their use requires understanding of the display properties.

PP-CAT-042

Pharmacological intervention to improve distance and near vision in uncomplicated pseudophakia

Claes Feinbaum[1], Franci Salamun[2], Katja Matovic[2], Sudi Patel[1]
[1]*Vision4You, Lund, Sweden,* [2]*Medicinski Center Vid, Slovenia, Nova Gorica, Slovenia*

To determine if (i) uncorrected distance (UDVA) and near acuity (UNVA) and, (ii) clinical depth of field (DoF) can be improved by topical administration of drops inducing changes in pupil size in pseudophakic eyes with low distance prescriptions.
Setting: Medicinski Center Vid, Slovenia & Vision4You, Lund, Sweden.
Methods: UDVA was measured using a standard LogMAR chart, UNVA was measured using standard near chart at 40 cm. DoF, at distance was measured by asking the patient to look at smallest letter with the best correction. Firstly, plus (a) lenses were racked up on the refractor head until the patient reported quality of vision was unacceptable. This was repeated using minus lenses (b). Irrespective of lens sign, DoF = a + b. Mesopic pupil size was measured by infrared pupillometry. Two drops of a custom made topical medication (consisting of an NSAID, an ocular lubricant and parasympathomimetic) were instilled and measurements were repeated 30–45 min later. All subjects were complication free pseudophakes (33 eyes). All acuity values were converted to decimal notation.
Results: Reporting just the significant association ($p \leq 0.001$) encountered in the 33 cases (eyes). The mean (± SD) before (i) and after (ii) findings were

1 Depth of Field (i) 1.546 D (0.49), (ii) 2.62 D (0.49)
2 Pupil size (i) 3.11 mm (0.47), (ii) 1.98 mm (0.28)
3 UDVA (i) 0.68 (0.17), (ii) 0.93 (0.18)
4 UNVA (i) 0.34 (0.18), (ii) 0.73 (0.14)

Conclusion: In this cohort of cases, both near and distance UDVA improved after drop instillation. This was equivalent to one line of improvement. UNVA also improved and converting to Jaeger categorisation, the results show that the intervention improved near vision from J6 to J3 in some cases. A 1 mm reduction in pupil size was associated with an increase in depth of field of 1.0 D in the pseudophakic eyes.

PP-CAT-061

Evaluation of a new multifocal intraocular lens with a lower near addition

Florian Kretz, Mary Safwat, Katharina Linz, Anna Fitting, Ramin Khoramnia, Tamer Tandogan, Gerd Auffarth
International Vision Correcion and Research Centre (IVCRC) & David J Apple Laoratory, Department of Ophthalology, University of Heidelberg, Heidelberg, Germany

Purpose: Clinical evaluation of two diffractive, pupil independent, multifocal intraocular lenses (MIOL) with a different near addition (Tecnis ZKB00 & ZLB00, AMO, USA).
Methods: The ZKB00 and the ZLB00 MIOL with a near addition of + 2.75 D and + 3.25 D were implanted during cataract surgery and refractive lens exchange. Follow up examinations included: monocular and binocular uncorrected distance visual acuity (UDVA), corrected distance visual acuity (CDVA), uncorrected near visual acuity (UNVA), distance corrected near visual acuity (DCNVA), corrected near visual acuity (CNVA) and individual reading distance and speed (Salzburg Reading desk).
Results: One month after surgery, the median monocular UDVA was 0.1 [logMAR]. Postoperative median monocular UNVA was 0.15 and binocular UNVA was 0.05. The patients were very satisfied with their outcome.
Conclusions: The ZKB00 and ZLB00 MIOL with a lower near addition provide good functional results after surgery with a high percentage of spectacle independence and patient satisfaction. Especially the comparison of the individual reading distance and speed will be very interesting at the 3 month visit.

PP-COR-017

Prevalence of meibomian gland disorder in a cohort of Norwegian dry eye disease patients

Reza A Badian[1,2], Jon R Eidet[2,3], Sten Rìder[2,3], Øygunn A Utheim[2,4], Xiangjun Chen[5], Aleksander Stojanovic[3,6], Tor P Utheim[2,5]
[1]Department of Ophthalmology, Innlandet Hospital Trust, Elverum, Norway, [2]The Norwegian Dry Eye Clinic, Oslo, Norway, [3]SynsLaser Kirurgi, Tromsoe and Oslo, Norway, [4]Department of Ophthalmology, Oslo University Hospital, Oslo, Norway, [5]Unit of Regenerative Medicine, Department of Medical Biochemistry, Oslo University Hospital, Oslo, Norway, [6]Department of Ophthalmology, University Hospital of North Norway, Tromsoe, Norway

Purpose: To assess symptoms and signs in dry eye disease (DED) patients with and without meibomian gland disorder (MGD).
Methods: DED patients with and without MGD were consecutively included in the study. All patients received an extensive ophthalmological work-up and completed two self-assessment questionnaires: McMonnies questionnaire (MDEIS) and Ocular surface disease index (OSDI).
Results: Three hundred and forty seven patients (50% women and 50% men) with DED of different etiologies were included. Of all DED patients, 9% (30 patients) showed no signs of MGD (i.e., normal meibum expressibility and quality). Patients with or without signs of MGD did not differ from each other with respect to age ($p = 0.477$) nor in the number of systemic prescription drugs used (0.198). Signs of MGD were more prevalent in males (95%) than in females (88%; $p = 0.025$). After controlling for gender effects, patients with signs of MGD had a higher MDEIS ($p = 0.012$), but not OSDI ($p = 0.371$), score than those without MGD.
Conclusions: The majority of DED patients, in particular males, showed some degree of MGD. In our cohort, patients with MGD scored higher than patients without MGD on MDEIS, but not on OSDI.

PP-COR-018

Contact lens use and dry eye disease

Reza A Badian[1,2], Tor P Utheim[2,3], Sten Rìder[2,4], Øygunn A Utheim[2,5], Xiangjun Chen[3], Aleksandar Stojanovic[4,6], Jon R Eidet[2,4]
[1]Department of Ophthalmology, Innlandet Hospital Trust, Elverum, Norway, [2]The Norwegian Dry Eye Clinic, Oslo, Norway, [3]Unit of Regenerative Medicine, Department of Medical Biochemistry, Oslo University Hospital, Oslo, Norway, [4]SynsLaser Kirurgi, Tromsoe and Oslo, Norway, [5]Department of Ophthalmology, Oslo University Hospital, Oslo, Norway, [6]Department of Ophthalmology, University Hospital of North Norway, Tromsoe, Norway,

Purpose: To assess possible adverse effects of contact lens use in a cohort of dry eye disease (DED) patients.
Methods: DED patients either with or without contact lens use were consecutively included in the study. All patients received an extensive ophthalmological work-up and completed the Ocular Surface Disease Index (OSDI) questionnaire.
Results: Three hundred and thirteen patients with DED of different etiologies were included. 8% (24 patients) of the cohort used contact lenses, where of 22 patients used soft contact lenses, and the remaining, hard contact lenses. After controlling for gender, age and the number of systemic prescription drugs used, patients using contact lenses had a lower OSDI score (9.9 ± 8.7) than non-contact lens wearers (14.5 ± 8.1) ($p < 0.5$). Significant reduction (53.8 ± 1.8) of corneal sensitivity measured with the Cochet-Bonnet esthesiometer was observed in patients wearing hard contact lenses compared to soft contact lens users (60.0 ± 0.0), and those not using contact lenses (59.2 ± 3.1) ($p < 0.5$). No other differences between contact lens and non-contact lens users could be observed in our cohort.
Conclusions: Contact lens users have a lower OSDI score than non-contact lens users. Unexpectedly, contact lens users as a group (including both soft and hard contact lenses) did not show any other differences compared to non-contact lens users.

PP-COR-019

A novel proteotoxic stress associated mechanism for macular corneal dystrophy

Szabolcs Felszeghy[1], Eszter Szalai[2], Lászlo Módis Jr[2], András Berta[2], Adrian Smedowski[3], Zoltán Hegyi[1], Niko Kivinen[4], Johanna Viiri[4], Bogumil Wowra[3], Giedrius Kalesnykas[4], Dariusz Dobrowolski[4], Edward Wylegala[3], **Kai Kaarniranta**[4,5]
[1]Department of Anatomy, Histology and Embryology, University of Debrecen, Medical and Health Science Centre, Debrecen, Hungary, [2]Department of Ophthalmology, University of Debrecen, Medical and Health Science Centre, Debrecen, Hungary, [3]Ophthalmology Clinic, District Railway Hospital, Katowice, Poland, [4]Department of Ophthalmology, Institute of Clinical Medicine, University of Eastern Finland, Kuopio, Finland, [5]Department of Ophthalmology, Kuopio University Hospital, Kuopio, Finland

Propose: Macular corneal dystrophy is an autosomal recessive disorder. It has been estimated that a major part of cellular proteolysis occurs in proteasomes, but the importance of proteasomes and the other proteolytic pathways including autophagy in aging cells is poorly known. Prior to proteolysis, heat shock proteins (Hsps), agents that function as molecular chaperones, attempt to refold misfolded proteins and thus prevent the accumulation of cytoplasmic protein aggregates. There is intriguing evidence suggesting that protein SQSTM1/p62, together with autophagy, has a role in the pathology of different degenerative diseases. The aim of our study was to analyse the distribution pattern and the expression levels of Hsp70, SQSTM1/p62 and ubiquitin protein conjugates in corneal buttons from human malucular dystrophy patients and in HCE-2 cells under proteasome inhibition *in vitro* conditions.
Methods: Four cases of macular dystrophy and 4 normal human corneal buttons were examined for their expression patterns of Hsp70, SQSTM1/p62 and ubiquitin protein conjugates by fluorescent immunohistochemistry. Expression levels in response to different concentrations of proteasomal inhibitor treatment (MG-132) were also analysed in HCE-2 cells by western blotting and transmission electron microscopy.
Results: Highly elevated Hsp70, SQSTM1/p62 and ubiquitin protein conjugates were observed in intracellular space of the epithelial cells in macular dystrophies. All the studied proteins were also highly elevated under proteasome inhibition in human corneal epithelial HCE-2 cells in cell cultures. We demonstrated the upregulation of Hsp70, SQSTM1/p62 and ubiquitin protein conjugates in the basal epithelial cells and stromal keratocytes in macular dystrophy.
Conclusions: This unique study first reveals a novel mechanism in macular dystrophy pathology involved in impaired proteosomal proteolysis. This work was financially supported by grants from Academy of Finland, the Finnish Eye Foundation, and John von Neumann Fellowship and by the grant of RH/885/2013.

PP-COR-064

Making the transition from DSEK to DMEK. Experiences during the learning curve

Eydís Ólafsdóttir, Marios Panagiotopoulos
Örebro University Hospital, Örebro, Sweden

Purpose: To report our early experiences with DMEK (Descemet's Membrane Endothelial Keratoplasty).
Methods: All patients undergoing DMEK during the first year were included.
Results: Visual results and complications from our first series of cases will be presented.
Conclusion: DMEK surgery is more technically demanding than DSEK, but the visual results are very promising.

PP-COR-065

Inverse Femtosecond laser DSEK

Marios Panagiotopoulos, Eydís Ólafsdóttir
Örebro University Hospital, Örebro, Sweden

Purpose: To present the results of the first inverse Femtosecond laser DSEK operations at our department.
Methods: Patients with Fuch's dystrophy and bullous keratopathy were included. Exclusion criteria were previous penetrating keratoplasty, glaucoma, ectatic corneal disease and aphakia. Donor lamella preparation was done with the Ziemer Z6 Femtosecond laser in lamellar keratoplasty mode, from the endothelial side using an artificial anterior chamber specially adapted for the laser.
Results: Visual results and postoperative endothelial cell counts will be presented.
Conclusion: The Ziemer Z6 femtosecond laser seems to be a safe tool to prepair donor tissue for DSEK.

PP-COR-073

One year follow-up after dry eye treatment according to the International Dry Eye Workshop (2007)

Sten Raeder[1,2], Tor Paaske Utheim[1,3], Øygunn Aass Utheim[1,4], Xiangjun Chen[1,2], Aleksandar Stojanovic[2,5], Jon Roger Eidet[1,3], **Reza A Badian**[1]
[1]The Norwegian Dry Eye Clinic, Oslo, Norway, [2]SynsLaser Kirurgi, Tromsoe and Oslo, Norway, [3]Unit of Regenerative Medicine, Department of Medical Biochemistry, Oslo University Hospital, Oslo, Norway, [4]Department of Ophthalmology, Oslo University Hospital, Oslo, Norway, [5]Department of Ophthalmology, University Hospital of North Norway, Tromsoe, Norway

Purpose: To evaluate the preliminary effects of treatment of dry eye disease at one year follow-up according to the guidelines given by the Tear Film and Ocular Surface Society (The International Dry Eye Workshop (2007).
Methods: All patients received an extensive ophthalmological work-up including screening for subjective symptoms, conjunctival injection, fluorescein vital staining, tear film break up-time (TFBUT), and Schirmer's test. Disease severity was graded according to dry eye severity level (DESL level 1–4), and treatment was based on DESL.
Results: Twenty-eight patients with DED of different etiologies were included. Vital staining dropped sharply from 2.0 ± 2.1 before treatment to 0.7 ± 0.64 after treatment ($p < 0.05$). TFBUT increased from 5.8 ± 4.1 before treatment to 8.3 ± 4.9 after treatment ($p < 0.05$). Dry eye severity level (DESL) prior to and after treatment was 2.00 ± 0.6 and 1.76 ± 0.64, respectively ($p = 0.2$).
Conclusions: Treatment of dry eye disease results in significantly lower vital staining score and significantly higher TFBUT one year after initiation of treatment. DESL was reduced, but did not reach significance.

PP-COR-078

Case report: Bacterial keratitis with an unusual and elusive etiologi: Nocardia beijingensis

Björn Johansson[1], Per Fagerholm[1], Margareta Claesson Armitage[2], Ulf Stenevi[2], Gabor Petranyi[3], Neil Lagali[1]
[1]Department of Ophthalmology and Department of Clinical and Experimental Medicine, Linköping University, Linköping, Sweden, [2]Department of Ophthalmology, Sahlgrenska University Hospital, Mölndal, Göteborg, Sweden, [3]Department of Clinical Microbiology, Linköping University Hospital, Linköping, Sweden

Purpose: To describe diagnostic and therapeutic difficulties encountered in a case of keratitis caused by Nocardia beijingensis
Methods: Patient history, laboratory reports, and photographic documentation were extracted from digital patient record files at Linköping and Sahlgrenska University Hospitals.
Results: A 21-year old man developed progressive superficial keratitis in the right eye after a journey to Sri Lanka. At first presentation visual acuity was 1.0 (20/20) and a small central corneal superficial lesion 0.5×0.5 mm was found. Initial antibacterial treatment was ineffective, and the lesion increased in size rapidly, covering the upper half of cornea. Visual acuity deteriorated down to finger counting. Repeated cultures from corneal scrapings did not reveal any bacterial or fungal etiology. The history together with clinical course and appearance of the lesion suggested Nocardia keratitis, and treatment with amikacin topically was started, still with no improvement. Confocal microscopy showed filamentary structures consistent with Nocardia. Extended culture from new scrapings and a corneal biopsy finally revealed Nocardia beijingensis, resistant to amikacin. Condition improved after treatment was changed to trimetoprim-sulfametoxazol orally and topically. After healing, central scarring remains, limiting visual acuity to 0.2.
Conclusions: This case demonstrates difficulties diagnosing unusual etiology of bacterial keratitis. Methods and strategies for correct diagnosis and treatment in keratitis cases with unclear etiology include repeated cultures from scrapings and biopsies, in direct communication with microbiological expertise. Long culture times may be necessary to detect slowly growing organisms. Confocal microscopy and histology may give new information if cultures are inconclusive or negative. Even after failing initial treatment the clinical picture together with a meticulous history taking may provide important clues to the diagnosis.

PP-COR-087

Evaluation of our experience with ReLEx SMILE technique in the first 1000 myopic eyes

Robert Kadlec, Richard Havranek, Jana Kacerovska, **Zuzana Mlynkova**
Ocni klinika Horni Pocernice, Prague, Czech Republic

The authors evaluate the latest method of laser reffractive surgery using only femtosecond laser inside the corneal stroma.

ReLEx SMILE method was introduced in Horní Počernice Eye Clinic in February 2012 and to date we have operated on more than 1000 eyes. The average age of patients is 31 years and the most common removed diopter range is from −2.0 to −8.0 D spherical equivalent.

The authors point out that this technique doesn't interfere with the integrity of the surface portion of the cornea and it isn't necessary create the flap.

Intraoperative complications mainly lie in the possible erosion of the epithelium at sidecut, postoperative complications are significantly smaller than femtoLASIK, because there are no complications associated with the creation of a flap. Simultaneously, the method is safer and more applicable for thinner corneas, due to stability of the Bowmans membrane.

PP-COR-089

Sports-related eye trauma

Tiina Leivo, Anna-Kaisa Haavisto, Ahmad Sahraravand
Helsinki University Eye Hospital, Helsinki, Finland

Purpose: To present the epidemiology, treatment and outcome of sports-related eye injuries. To give recommendations for the use of protective eye-wear.
Methods: All new sports-related eye injury patients in Helsinki University Eye Hospital in one year (2011–2012). Epidemiological, clinical and resource use data from the hospital records, completed by a patient questionnaire. Follow-up 3 months. Statistical analysis of the data.
Results: One hundred and forty-nine 149 sports injury patients. Floorball caused 47, football 19, tennis 15, ice-hockey 12, cycling 8, badminton 7, basketball, baseball and combat sports 6 eye traumas.

Relative to sports participation rate, injuries were commonest in rink bandy, floorball and tennis.

Contusion was the primary diagnosis in 114 cases, followed by lid trauma, orbital trauma and penetrating injury. 32% of contusion patients had severe secondary diagnoses.

Number of outpatient visits was 459, inpatient days 25. 23 patients needed major, 12 minor surgery. Sick leave days were 1220, physical activity restriction 2982.

Seventeen patients had permanent functional impairment. Relative to the number of injuries, a permanent impairment was significantly commonest in ice-hockey. 18 cases were in OTS classes 1–3. 7 patients used protective eye-wear, 4 in ice-hockey.

Compared to previous data, the number of eye injuries had statistically significantly declined in floorball, badminton, rink bandy and squash, increased in tennis and ice-hockey. In floorball in under 14 years' age group, where protective eye-wear had meanwhile been made mandatory; 11 (2002–03) vs 1 (current) eye trauma was observed.
Conclusions: Floorball is the leading eye injury causing sport in Finland. Protective eye- wear in junior floorball has been effective in preventing injuries and should be mandatory in all age groups. In ice-hockey eye injuries are in incline, relatively severe and a third of injuries incurred in spite of eye mask. The proper use of the mask should be emphasised and the masks further developed.

PP-EDU-003

To see or to be seen: On the development of the white cane

Sibylle Scholtz[1], Stefanie Liebers[2], Florian Kretz[1], Gerd Auffarth[1]
[1]International Vision Correction Research Centre (IVCRC), Department of Ophthalmology, University of Heidelberg, Germany, Heidelberg, Germany, [2]Schloss-Schule Ilvesheim – Schule für Blinde und Sehbehinderte, Ilvesheim, Germany

Purpose: Blind people have used canes as mobility tools for centuries. The invention of the 'White Cane' seems to be a story of controversies. This poster will reflect the different starting points of the origin(s) of the White Cane.
Methods: Selective literature research of books and articles in journals via PubMed, Google Scholar and Google.
Results: Three people played the mayor role in the history of the white cane: James Biggs (UK), 1921, who became blind after an accident, he painted his walking stick white to be seen better. Guilly d'Herbemont (France), 1931, who gave the first white canes to blind people in France. Member of Lions Club George A. Bonham (USA), 1930, saw the danger in using black canes as well, 1931 the Lions Club International begun promoting the use of white canes for the blind successfully in the US.
Conclusions: Today the white cane is not only a mobility device, it is as well a symbol for blind people and a Medical Device class I. To keep the awareness high for blind and visually impaired people the US proclaimed October, 15. As 'White Cane Safety Day' in 1964.

PP-EDU-005

A cold nose will show you the way: On the history of guide dogs for the blind

Sibylle Scholtz[1], Christine Kobelt[2], Andreas Schmelt[3], Florian Kretz[1], Gerd Auffarth[1]
[1]International Vision Correction Research Centre (IVCRC), Department of Ophthalmology, University of Heidelberg, Germany, [2]Tierarztpraxis Dr. Kobelt, Karlsruhe, Germany, [3]Deutsche Blindenführhunde e.V., Berlin, Germany

Purpose: Their job is not limited to a schedule from 9 to 5, but they don't receive any salary. They can distinguish between right and left, are intelligent with high work ethics, highly able to withstand stress but they get vaccinated, castrated and microchips-implanted. This poster will briefly reflect the major steps of the history of guide dogs for the blind.
Methods: Selective literature research of books and journal articles via PubMed, Google Scholar and Google, close cooperation with German Guide Dogs Association (Deutsche Blindenführhunde e.V.).
Results: In Antiquity dogs shown together with blind persons on paintings can be understood as their companions, not as guide dogs. First ideas using them for guiding blind people arose in Europe in 1780 in Paris (France) and in 1788 in Vienna (Austria). First training instructions for these dogs have been established in Vienna in 1819. Due to the high number of war-blind persons after WWI the interest in guide dogs was high. The first school for guide dogs for the blind worldwide was founded in 1916 in Oldenburg, Germany. Dorothy Harrison Eustis, American, being highly impressed by the benefits of guide dogs, founded a school for trainers of guide dogs in Switzerland in 1927. The same year she wrote an article on this topic which was published in America's 'Saturday Evening Post'. This article contributed to the international breakthrough of using guide dogs for the blind.
Conclusions: Today guide dogs for the blind are widely accepted as therapeutic means for blind people and are mostly supported by the local Government and/or health insurances. They offer a great source of independence to their users. Above that dogs are one of life's noble gifts, they are companions and a wet-nosed protection shield – they make life's journey not only possible but also more worthwhile.

PP-EDU-028

The slit lamp – Report back to Allvar Gullstrand

Marcus-Matthias Gellrich
Ophthalmological Practice, Kellinghusen, Germany

Purpose: In 1911 Allvar Gullstrand introduced the 'Nernst Spaltlampe' which thereafter became the most important instrument in ophthalmology – the slit lamp. Outstanding ophthalmologists – among them Alfred Vogt and Hans Goldmann – together with gifted engineers successfully applied the technical knowledge at that time. The author makes suggestions to broaden the spectrum and accuracy of today's slit lamp diagnostics.
Methods: Over a period of fourteen years (2000–2014) concepts were worked out in the clinical routine of an ophthalmological practice with the aim to monitor any disease in ophthalmology – using a Zeiss video slit lamp SL 105. Altogether 130 700 slit lamp images were taken in 9200 patients.
Results: The following actions resulted in applications which can be useful in clinical work with a video slit lamp action – effect minus lens in front of slit lamp objective – portrait and squint images image reconstruction software (Hugin) – fundus overviews flicker test (Power Point) – visualizing small changes (e.g. glaucoma) analysis of videosequence – documentation of RAPD analysis of Purkinje images – power of intraocular lenses reproducible examination conditions – pachymetry
Conclusions: With a sea change from the principles of biomicroscopy to videography the slit lamp could regain a central role as an affordable

instrument for imaging of nearly any disease in modern ophthalmology. As a proof for that the author published at the beginning of 2014 the first 'videographic' slit lamp atlas covering the whole range of ophthalmology (The slit lamp – applications for biomicroscopy and videography, Springer).

PP-EDU-044

Simulation-based certification for cataract surgery

Ann Sofia Skou Thomsen[1,2], Hadi KjÌrbo[1], Morten Dornonville de la Cour[1], Lars Konge[2]
[1]Department of Ophthalmology, Glostrup Hospital, Capital Region, Denmark, [2]Centre for Clinical Education, Rigshospitalet, Copenhagen, Denmark

Purpose: To evaluate the cataract surgical modules of the EyeSi™ simulator in order to assess the progression in surgical expertise. In this sub-study we have defined an expertise level which qualifies the student to progress into *in vivo* surgery.
Methods: Thirteen of a total of 14 modules were found to have content validity and were therefore included in the assessment. Difficulty levels of the modules were defined by evaluating previous validation studies on the EyeSi simulator. We included 26 residents in ophthalmology (no cataract surgical experience) and 11 experienced surgeons (> 4000 cataract surgeries) in the study. All subjects completed all 13 modules twice.
Results: Total module score on 8 out of 13 modules showed significant discriminative ability between the groups, and a reliability coefficient of 0.773 ($p < 0.001$). A pass/fail level was defined from the total score on these 8 modules using the Contrasting Groups Method. As a consequence, 10 of 11 experienced surgeons (91%) and 4 of 26 novices (15%) passed the test. To ensure a sufficient reliability, competency level has to be reached in two consecutive tests. Learning curves of two novices showed that 10 repetitions (approximately 16 hr of training) were needed before reaching the defined expertise level in two consecutive tests.
Conclusion: A performance test, consisting of 8 modules on the EyeSi simulator, is a useful and reliable tool for assessment of cataract surgical expertise. The defined expertise level has to be reached in two consecutive tests before the student is qualified to progress into supervised *in vivo* surgery.

PP-GLA-016

Ocular surface disease signs among glaucoma patients

Minna Parkkari[1], Auli Ropo[2], Hannu Uusitalo[1,3]
[1]Department of Ophthalmology, SILK, University of Tampere, Tampere, Finland, [2]Santen Oy, Tampere, Finland, [3]Tauh Eye Center, Tampere, Finland

Purpose: The objective of this study was to evaluate prevalence of ocular surface disease (OSD) signs among glaucoma outpatients.
Methods: The study was a multicentre, cross-sectional, single visit study in which 564 glaucoma patients who were using intraocular pressure (IOP) lowering medications and 51 untreated control patients attended private ophthalmologist offices in Finland. Ophthalmological assessments for signs of OSD included examination of the: 1. eyelids (redness severity scale none, mild, moderate, severe), 2. evaluation of conjunctival redness (reference photos, scale 0–4), 3. corneal fluorescein staining, Oxford scale 0-V each, 4. nasal and temporal conjunctival fluorescein staining (Oxford scale 0-V each and 0-X combined) 5. measurement of tear fluid break-up time (in seconds) and 6. Shirmer's test (in millimeters).

The presence of ocular signs (zero to six) was evaluated from the both eyes of glaucoma and control patients and the worst eye values were recorded for further analyses. Abnormal ocular signs was determined as follows: eyelids redness: mild, moderate, severe; conjunctival redness: grade 2–4; corneal fluorescein staining: grade I-V; combined nasal and temporal conjunctival fluorescein staining: III-X; tear fluid break-up time: < 10 seconds and Shirmer's test: < 10 ml.
Results: One per cent of the glaucoma patients (7/564) had no abnormal ocular signs compared to 16% (8/51) of the control patients. Of the glaucoma patients, 93% had two or more individual abnormal ocular signs compared to 63% of control patients. Five abnormal signs was the most common category among glaucoma patients (26%) and 16% had 6 abnormal signs. Among the control subjects the corresponding figures were 6% and 0%.
Conclusion: According to this study the signs of OSD are more common in glaucoma patients compared to control patients without glaucoma medication.

PP-GLA-039

Surgical management in primary congenital glaucoma patients during a follow-up period from 3 to 34 years

Karen Bjerg Pedersen[1], Lisbeth Sandfeld[2], Nathalie Zibrandtsen[3], Daniella Bach-Holm[1]
[1]University Hospital of Copenhagen, Glostrup, Denmark, [2]Department of Ophthalmology, Roskilde Hospital, Roskilde, Denmark, [3]Møller & Zibrandtsen Eye Clinic, Lyngby, Denmark

Purpose: To quantitatively describe the surgical management in patients with primary congenital glaucoma (PCG) in a 3 to 34 year follow-up study.
Methods: Fifty-four patients out of 88 patients diagnosed with PCG in Denmark from 1977–2006 participated in a follow-up examination performed at the Department of Ophthalmology, University Hospital of Copenhagen, Glostrup in 2009–2011. Main surgical procedures performed were trabeculotomy, trabeculectomy and trans-scleral cyclodiode laser. In patients with bilateral PCG, the eyes were divided into first and second eye.
Results: Ninety-two eyes with PCG were included. Thirty-eight patients (70.4%) had bilateral and 16 (29.6%) unilateral PCG. In patients with unilateral PCG, the initial surgical procedure was trabeculotomy in 14 eyes and trabeculectomy in 1 eye. In patients with bilateral PCG (first/second eye), 30/27 eyes underwent trabeculotomy and 7/6 eyes trabeculectomy as initial surgical procedure. Median age at initial surgery was 5.0 months (range 1 week to 108 months). For patients with bilateral PCG, median time between initial surgery of first and second eye was 1.0 month (range 0–84 months). More than one surgical procedure was required in 18.8% of patients with unilateral PCG compared to 36.8%/39.4% (first/second eye) in patients with bilateral PCG ($p = 0.5$). The difference in total number of surgeries in patients with unilateral PCG and bilateral (first/second) PCG was not statistically significant ($p = 0.1$).
Conclusion: The initial surgical procedure of choice was trabeculotomy in the majority of PCG patients. One fifth of unilateral and more than one third of bilateral PCGs needed more than one surgical procedure. Thus it seems more difficult to get satisfactory primary surgical results in patients with bilateral PCG compared to patients with unilateral PCG though the results were not statistically significant.

PP-ONC-015

Regression of choroidal metastases after systemic chemotherapy

Nina Krøll Angelsen[1], Kristian Fossen[1], Nils Eide[2]
[1]Universitetssykehuset Nord Norge, Tromsø, Norway, [2]Oslo universitetssykehus, Oslo, Norway

Purpose: We present three patients with choroidal metastasis responding to systemic chemotherapy.

Methods: Case reports of three patients with choroidal metastasis. A 70 year old male with oesophageal cancer, a 43 year old woman with breast cancer and a 59 year old male with lung cancer. All these patients received systemic chemotherapy only as first-line treatment.
Results: All three patients showed primary regression of choroidal metastasis after systemic chemotherapy. Two of the patients were treated only with systemic therapy, as the effect of chemotherapy until now made local ocular treatment unnecessary. The third case, a 59 year old male with lung cancer showed marked regression of the choroidal metastasis after the first cycle of carboplatin and vinorelbine. Due to haematological toxicity, the chemotherapy dosage was reduced in the three subsequent cycles. Progression of primary tumour and choroidal metastasis was seen after the fourth cycle. External beam radiation was then chosen as second line treatment.
Conclusions: These cases demonstrate that regression of choroidal metastasis after systemic chemotherapy can occur. This is consistent with earlier published case reports. Patients with metastasis in multiple organs of known origin with simultaneously choroidal metastases can be treated with systemic therapy as first-line treatment. The ophthalmologist should monitor them closely as progression of ocular metastasis could indicate need of local treatment. External beam radiation therapy, plaque therapy and other local treatment are options when choroidal metastases do not respond to chemotherapy, in cases of local relapse during treatment with systemic chemotherapy or when tumour origin is unknown.

PP-ONC-026

The correlation between survival and AJCC staging and genetic status in uveal melanoma patients

Mette Bagger[1,2], Morten T Andersen[2], Steffen Heegaard[3,4], Mette K Andersen[2], Jens F Kiilgaard[1,4]
[1]Department of Ophthalmology, Rigshospitalet, Copenhagen, Denmark, [2]Department of Clinical Genetics, Rigshospitalet, Copenhagen, Denmark, [3]Institute of Eye Pathology, Copenhagen University, Copenhagen, Denmark, [4]Department of Ophthalmology, Glostrup Hospital, Glostrup, Denmark

Purpose: To evaluate the predictive value of genetic status and AJCC (American Joint Committee on Cancer) staging in patients with uveal melanoma.
Methods: Patient charts for patients treated from 2009 through to 2012 were reviewed. Information on survival, AJCC stage and chromosomal status were collected.
Results: A total number of 151 patients with either ciliary or choroidal melanoma were treated at Copenhagen University Hospital from 1 January 2009 through to 31 December 2012. In 12 cases no genetic information were found. 27.5% of the patients had a normal chromosome status. Normal chromosomal status was found in 9 of 26 stage 1, 28 of 91 stage 2, 4 of 32 stage 3 and 0 of 2 stage 4 patients. An overall survival rate of 63% after 4 years was observed. Only 1 melanoma related death was observed in the group with normal chromosomal status. There was a direct correlation between AJCC staging and death and number of abnormal chromosomes and death. A combination of AJCC staging and chromosomal status gave the best prediction of melanoma related death.
Conclusion: AJCC staging and chromosomal status are good predictors of survival. For prediction of melanoma related death a combination of AJCC staging and chromosomal status gave the best results.

PP-PAE-013

Ophthalmological findings and sequels after different types of paediatric CNS tumours

Ulrika Lidén, Agneta Rydberg, Lena Falkman, Ingrid Axelsson, Kerstin Hellgren
Astrid Lindgrens Childrens Hospital Karolinska Institutet, Stockholm, Sweden

Introduction: Presenting early ophthalmological symptoms, long term sequels and rehabilitation strategies in a cohort of paediatric CNS tumours at a paediatric university hospital.
Purpose: To evaluate the frequency of ophthalmological disturbances related to paediatric CNS tumours: To illustrate ophthalmological findings in different locations and stages of paediatric CNS tumours and to emphasize the importance of visual follow up and rehabilitation.
Material and Methods: During a three year period we consecutively included all children referred to the paediatric neuroophthalmological clinic at Astrid Lindgrens children's hospital with diagnosed CNS tumours. A thorough age relevant clinical examination was done by the authors including visual fields, ocular motility and fundus imaging when possible.
Results: A group of 78 children with brain tumours were included in the project. At the time of referral ophthalmological findings were common, various and often previously missed in this group of children. During follow up, ophthalmological findings were often used for timing of tumour treatment. Different rehabilitation strategies were applied, depending on the symptoms.
Conclusions: The ophthalmologist plays an important role in detecting early symptoms leading to early diagnosis and treatment in many types of paediatric CNS tumours. The role in rehabilitation of residual visual problems and detecting worsening of visual status is equally important.

PP-RET-006

Characteristics of retinal nerve fiber layer in severe preterm Malay children

Alshaarawi Salem[1,2]
[1]Universiti Sains Malaysia, Kubang Kerian, Malaysia, [2]Islamic University, Gaza, Occupied Palestinian Territory

Introduction: Optic nerve anomalies are unknown ophthalmic diseases in prematurity children. Better knowledge of the retinal characteristics of the preterm children would permit more accurate diagnosis and treatment of anomalies that are associated with preterm birth.
Purpose: To examine retinal nerve fiber layer thickness in severe preterm Malay children aged 8–16 years old.
Methods: Total 32 children who were born at 32 weeks of gestation or less and 32 term children underwent imaging of retinal nerve fiber layer thickness with the HRT3 (Heidelberg Engineering, Germany). HRT data also were analysed using version 3 of the software to obtain thickness measurements for 6 sectors (Temporal quadrant, temporal superior, temporal inferior, nasal quadrant, nasal superior, and nasal/inferior). Retinopathy of prematurity and amblyopic children were excluded from this study. Statistical analysis were conducted using PASW Statistics version 18.
Results: Peripapillary retinal nerve fiber layer in nasal quadrant, nasal/superior quadrant, and nasal/inferior quadrant were statistically thinner in severe preterm children compared with term children ($p < 0.05$). However, there was no statistically significant difference between the two groups in the temporal quadrant, temporal superior, and temporal inferior ($p > 0.05$).
Conclusion: The significantly thinner retinal nerve fiber layer nasal side for severe preterm children suggests that prematurity is associated with subclinical optic nerve anomalies

PP-RET-038

The relationship of the results between multifocal-ERG and optical coherence tomography in the diagnosis and monitoring patients with age-related macular degeneration

Khaled Ait Ahmed, Tatyana Kamenskih
Saratov Stat Medical University, Saratov, Russia

Purpose: The study of the relationship (correlation) between the results of the macular area responses by multifocal-electroretinography (mf-ERG) and the retina's thickness obtained by the spectralis optical coherence tomography (OCT) in patients with age-related macular degeneration (AMD).

Material and Methods: Thirty patients (32 eyes) were enrolled in the study, the male/female's ratio was 1:1, with average age 65 ± 15 years old and with a visual acuity not less than 1/10 with maximal optical correction. Two groups were formed, from the first (experimental) group with 20 patients (22 eyes) suffering from AMD with sub-fovea localization were derived 2 sub-groups: 13 eyes with 'wet' form AMD, 9 eyes with 'dry' form AMD. The second (control) group was up to 10 patients without any AMD's lesions in retina. All patients were examined by OCT Spectralis which measured the retinal thickness of fovea and by mf-ERG which showed the amplitudes of bioelectrical answer of this area. All the results were analyzed and compared by the correlation statistical method with the calculation of the coefficient of correlation 'R'.

Results and Conclusions: Study's data showed a high level of negative correlation between the retinal thickness and electrical amplitudes of the fovea ($R = -0.93$ for 'wet' AMD group and $R = -0.84$ for 'dry' AMD group). This means that with the presence of AMD's lesions, the rise of retinal thickness (oedema, fibrosis) of the macular zone influences negatively on bioelectrical activity of this area. This fact is also confirmed by the low correlation of the control group ($R = -0.08$). In other words, the bioelectrical activity of the macular area of healthy persons without AMD is depending on other factors (age, gender, individual particularities and others). At last, mf-ERG in supplement of OCT can be used as an important element in complex exams and in the monitoring of patients with AMD.

PP-RET-052

Intravitreal Dexamethasone slow releasing device: next treatment of choice for persistent Diabetic Macular Oedema?

Pierfrancesco Mirabelli[1], Moa Eriksson[2]
[1]Department of Ophthalmology and Department of Clinical and Experimental Medicine, Linköping University, Linköping, Sweden, [2]Department of Ophthalmology and Department of Clinical and Experimental Medicine, Linköping University, Norrköping, Sweden

Purpose: Diabetic Macular Oedema (DME) is a leading cause of decreased vision in diabetic population. Laser photocoagulation has been the only available treatment for decades, until recent approval of anti-VEGF. The limitations of both therapies trigger new therapies to be found. Since inflammation plays a significant role in pathogenesis, dexamethasone given as intravitreal slow releasing device (Ozurdex) represents a possible new treatment. The aim of this study is to investigate the effects of Ozurdex on Best Corrected Visual Acuity (BCVA) and Central Retinal Thickness (CRT) in a group of patients with DME.

Methods: Retrospective study on 15 eyes of 12 patients with persistent DME, despite previous laser photocoagulation and/or anti-VEGF, treated with Ozurdex at Norrköpings Department of Ophthalmology during 2013 (9 males, 3 females; mean age: 66.6 years; mean duration of DME:3.93 years). BCVA ranged between ETDRS 36 and 70. 10 eyes had previously been treated with laser photocoagulation and 12 eyes with intravitreal anti-VEGF.

Results: Two months after Ozurdex injection, average BCVA gain was 4.4 letters ($p < 0.5$; range −4 to + 26), average CRT reduction 194.7 μm ($p < 0.001$; range 19–367), which represented a mean CRT reduction of 34.6% (range 4–62%). In 13 of 15 eyes, CRT decreased more than 10%.

Conclusions: In our patients, Ozurdex showed excellent CRT reduction 2 months after injection but only a modest effect on BCVA. The functional damage due to long-standing DME may represent a selection bias, since Ozurdex being an off-label treatment was offered as last option after failure of standard therapy. Despite the limitations due to retrospective design and small population size, our encouraging morphological results suggest that Ozurdex can be an effective treatment for DME, when given at an earlier stage.

PP-RET-058

The national database on diabetic retinopathy and maculopathy in Denmark

Nis Andersen[1], Jesper Hjortdal[2], Katja Cristina Schielke[2], Toke Bek[2], Majbrit Lind[2], Caroline Schmidt Laugesen[2], Henrik Lund-Andersen[2], Jesper Skov[1]
[1]Danish Ophthalmological Organizarion, Copenhagen, Denmark, [2]Danish Ophthalmological Society, Aarhus, Denmark

Purpose: To report data from the national database on diabetic retinopathy and maculopathy in Denmark.

Methods: Denmark with 5.5 million inhabitants has an estimated 300 000 diabetic patients, increasing by 27 000 newly diagnosed diabetic patients every year.

In 2012, 37 528 of these patients had more severe diabetes and were therefore seen in medical departments at various hospitals, while the majority of patients where regularly examined at their family doctor.

In 2012, 15 592 diabetic patients underwent regular screening for diabetic retinopathy and maculopathy in Danish eye departments, while the majority of patients were screened in ophthalmological practice.

All of these 15 592 diabetic patients where registered in the national database on diabetic retinopathy and maculopathy in Denmark termed DiaBase.

Results: Fourteen percent of 15 592 diabetic patients registered in the national database showed overall progression of retinopathy compared to 17% in 2011.

Three percent were refered for treatment for newly diagnosed diabetic macular oedema while 5% of the registered diabetic patients were refered to treatment for newly diagnosed proliferativ retinopathy. 1% of the diabetic patients are legally blind. Those numbers are stable in comparison to 2011.

Conclusions: The Danish National Database on diabetic retinopathy and maculopathy became nationwide in 2010. By collecting data from screening and treatment, the database is an excellent tool to observe prevalence and progression of diabetic eye disease in the future. Futhermore, data on patients who are screened in ophthalmological practice will be included from 2014.

PP-RET-059

Pharmacodynamics of Ranibizumab treatment of center-involved diabetic macular oedema

Delila Hodzic-Hadzibegovic[1,2], Birgit Agnes Sander[1], Henrik Lund-Andersen[1], Michael Larsen[1]
[1]Department of Ophthalmology, Glostrup Hospital, University of Copenhagen, Glostrup, Denmark, [2]Faculty of Health Sciences, University of Copenhagen, Copenhagen, Denmark

Purpose: To investigate if some eyes with center-involved diabetic macular oedema (DME) are undertreated by our current VEGF inhibitor regimen.
Methods: Retrospective analysis of 28 eyes in 20 patients with center-involved DME who underwent a spectral domain optical coherence tomography scan earlier than 4 weeks and at 4 weeks after most recent Ranibizumab injection for DME with central subfield thickness (CST) > 300 μm. Twelve eyes were treatment-naïve and 16 eyes were with recurrence of DME with a treatment free period of 3 months. The outcome parameter was presence or absence of > 10% reduction in foveal CST thickness at any time within 4 weeks after most recent Ranibizumab injection.
Result: Reduction in CST > 10% within 4 weeks was seen in 20 eyes in 16 patients. 11 eyes with reduced CST at 4 weeks showed a reduction in CST > 10% from 1 week after Ranibizumab injection. 5 eyes with reduced CST at 4 weeks showed a reduction in CST > 10% from 2 weeks. 2 eyes with reduced CST at 4 weeks showed a reduction in CST > 10% after 1 week and partial relapse beginning respectively at 2 and 3 weeks after Ranibizumab injection. 2 eyes demonstrated a reduction in CST that increased with time after injection. Absence of CST reduction was found in 8 eyes in 7 patients. No relation to age, diabetes duration, HbA1c and baseline CST.
Conclusion: No eye demonstrated a total loss of an early reduction in CST at 4 weeks. Early follow-up is warranted if follow-up after 4 weeks shows unsatisfactory status after VEGF inhibitor injection. Individual dose and retreatment interval titration may be attempted as a means of improving outcomes in VEGF inhibitor treatment of DME.

PP-RET-062

Fluocinolone acetonide intravitreal implant for diabetic macular oedema: a case report of a patient with a therapy resistant diabetic maculopathy

Florian Kretz, Gerd Auffarth, Katharina Linz
International Vision Correction and Research Centre (IVCRC) & David J Apple Laboratory, Department of Ophthalmology, University of Heidelberg, Heidelberg, Germany

Purpose: Evaluation of bilateral implantation of a Fluocinolone acetonide intravitrealimplant (Iluvien®) in a patient with a therapy resistant macular oedema.
Methods: An 87-year-old patient with a diabetic Retinopathy was treated consecutively for diabetic macular oedema within 5 years. The first consultation was in May 2009 presenting a diabetic maculopathy. In-between 2009 and 2013 both eyes were treated with focal laser photocoagulation and multiple intravitreal injections of Anti-VEGF and Triamcinolon. The following intravitreal injections were performed: on the left eye Triamcinolon ($n = 3$), Avastin ($n = 5$) and Lucentis ($n = 4$), on the right eye Triamcinolon ($n = 2$), Avastin ($n = 3$) and Lucentis ($n = 5$). Due to persistent diabetic macular oedema Iluvien was implanted on both eyes. Follow up examinations included visual acuity, OCT and intraocular pressure measurements.
Results: At 1 month after the implantation of Iluvien, uncorrected distance visual acuity (UDVA) increased to 0.58 logMAR compared to preoperative UDVA of 0.7 logMAR on the right eye. On the left eye UDVA increased to 0.9 logMAR compared to preoperative UDVA of 1.0 logMAR. In the 3-months follow-up visit the UDVA improved to 0.46 logMAR on the right eye and to 0.88 logMAR on the left eye. The intraocular pressure remained stable on both eyes at 13 and 12 mmHg compared to preoperative measurements. Retinal thickening at the center of the macular has been reduced on both eyes.
Conclusion: The implantation of Iluvien offers a sight-saving option for patients with chronic diabetic oedema who were previously treated with laser photocoagulation, intravitreal injections of Anti-VEGF or steroids without a sufficient response to the therapy. The long testing effect also reduces the amount of intravitreal injections and possible risks of complications.

PP-RET-069

Refractive error associated with age-related macular degeneration. Tromsø Eye Study

Maja Gran Erke[1], Geir Bertelsen[1,2], Tunde Peto[3]
[1]University Hospital of North Norway, Tromsø, Norway, [2]University of Tromsø, Tromsø, Norway, [3]NIHR Biomedical Research Centre at Moorfields Eye Hospital NHS Foundation Trust and UCL Institute of Ophthalmology, Head of Reading Centre, London, UK

Purpose: To examine associations between refractive error and age-related macular degeneration (AMD).
Methods: Population-based, cross-sectional study of Caucasians aged 65 to 87 years conducted in Norway in 2007/2008. Retinal photographs were graded for AMD. Late AMD was defined as geographic atrophy and/or neovascular AMD. Multivariable logistic regression analyses were performed based on autorefractor measurements and questionnaires addressing eye health, life style habits, and physical examination comprising anthropometric measurements and blood pressure.
Results: Gradable photographs were available for 4935 eyes/2514 participants with autorefractor measurements. We excluded 1090 eyes/561 participants from analysis due to cataract or missing data on cataract status, previous surgery for cataract, retinal detachment or refractive error. Late AMD was present in 58 of the remaining 3845 eyes (1.5%) or 46 of 1953 participants (2.4%). No high myopic (−6 or lower) eyes had late AMD.

In multivariable logistic regression analysis adjusted for age, sex, smoking, systolic blood pressure, body mass index, physical activity and cardiovascular disease we observed an association between spherical equivalent and late AMD when pooling right and left eyes together. We found that each dioptre increase in spherical equivalent in an eye was related to decreased odds of prevalent late AMD (odds ratio (OR) 0.86, 95% confidence interval 0.76–0.98, $p = 0.02$). These relationships did not reach statistical significance in analyses performed on right eyes or left eyes singly (OR 0.86, $p = 0.07$ and OR 0.86, $p = 0.2$ respectively). Further, in multivariable analysis of subjects classified based on the more severely affected eye, we observed no significant association between mean spherical equivalent from both eyes and late AMD (OR 0.91, $p = 0.3$).
Conclusions: Higher spherical equivalent was associated with lower prevalence of late AMD in the pooled analysis of right and left eyes controlled for confounders. This was not confirmed for analyses on participant level.

PP-RET-070

Optical coherence tomography grid decentration and its effect on macular thickness measurements

Khalil Ghasemi Falavarjani, Nasser Karimi, **Joobin Khadamy**, Foad Amirkourjani
Eye Research Center, Rassoul Akram Hospital, Iran university of medical science (IUMS), Theran, Iran

Purpose: To evaluate the rate of optical coherence tomography (OCT) grid decentration in a routine practice and its effect on thickness measurements.
Methods: During seven usual working days, spectral domain OCT 3D scans from various macular diseases including age related macular degeneration, diabetic macular oedema, macular pucker and central serous chorioretinopathywere studied. In each scan, the macular EDTRS grid was evaluated for decentration. After grid adjustment, the changes in central subfield thickness (CST) measurements were recorded.
Results: Ninety-three 3D scans of 93 eyes were evaluated. Thirty-three scans were unreliable for grid adjustment. In the remaining 60 eyes, grid decentration (> 1 microns) was found in 32 eyes (53.3%). The mean change in CST after grid adjustment was 41.1 ± 55.01 (3–251) microns. Clinically significant CST changes (> 8.5 microns) were found

in 24 eyes (40%). No significant relationship was found between underlying disease and corrected CST, and grid decentration.
Conclusion: Although the scan area should ideally be centered on the foveola, this may not occur in all patients. Considering that the grid is automatically registered on the scan area, if the scan is not centered on foveola, the thickness measurements on the grid is incorrect. The correction of the grid decentration is not routinely performed in busy clinical practices. Abnormal retinal thickness map was found in a significant number of scans. OCT 3D scans should be evaluated routinely for grid decentration.

PP-RET-071

Great real life improvement of treatment of neovascular Age-Related Macular Degeneration with few injections and reduced time from diagnosis to injection

Annette Rasmussen[1,2], Sara Brandi Bloch[1], Josefine Fuchs[1,2], Louise Hillerup Hansen[1], Michael Larsen[1,2], Henrik Lund-Andersen[1,2], Birgit Sander[1,2]
[1]Department of Ophthalmology, Glostrup Hospital, Glostrup, Denmark, [2]University of Copenhagen, Copenhagen, Denmark

Objective: To investigate visual baseline and outcome, treatment pattern, number of injections in four cohorts where the routine clinical treatment protocol was adapted during a six year period. Patients treated in a variable ranibizumab dosing regimen for neovascular age-related macular degeneration (AMD).
Design: Retrospective chart review.
Participants: One thousand hundred and eight patients/1185 eyes initiated intravitreal treatment with vascular endothelial growth factor inhibition for neovascular AMD the first 6 months of respectively 2007, 2009, 2011 and 2012.
Methods: Patient data from a database were retrieved from 2007 to 2013. Descriptive evaluation of main outcome measures for the cohort of patients. 1 year follow-up
Main Outcome Measures: Change in mean best corrected visual acuity (BCVA, Snellen), number of injections, causes of discontinuations, time to treatment, fellow eye.
Results: The mean age and BVCA at presentation were unchanged at 79 years and from 0.22–0.23 from 2007 to 2012. In 2007 and 2009 mean BCVA after 1 year was stable (0.21–0.25). In 2011 and 2012 mean BCVA had increased after 3 months (0.30; $p < 0.0001$) and after 1 year (in treatment 0.37–0.41; LOCF 0.29–0.31). The numbers of injections were 4.9 in 2007, 6.5 in 2009, 6.8 in 2011 and 6.8 in 2012. Time to treatment from diagnosis to the first injection was significantly ($p < 0.0001$) reduced from 2007 to 2012. Baseline BCVA in fellow eyes was unchanged, except in year 2007. In 2011 and 2012 after 1 year approximately 75% of patients were still in treatment, 10% had been discontinued because of apparent disease inactivity, 10% because of apparent lack of response and less than 5% mainly due to concurrent diseases and death.
Conclusion: Despite comparable age and baseline BCVA, outcome improved considerably with a stable number of first-year injections around 6.8, possibly because we switched to injecting on the same day the diagnosis was made.

PP-RET-092

Diabetic retinopathy in All Newly diagnosed Diabetic subjects in Skåne (ANDIS), Sweden, during 2008–2013. Baseline findings.

Carina Gustavsson[1], Mats Martinell[2], Mozhgan Dorkhan[3]
[1]Department of Clinical Sciences in Helsingborg/Ophthalmology, Helsingborg Hospital, Lund University, Helsingborg, Finland, [2]Department of Public Health and Caring Sciences, Family Medicine and Preventive Medicine, Uppsala University, Uppsala, Sweden, [3]Department of Clinical Sciences in Malmö/Endocrinology, Lund University, Lund, Sweden

Purpose: To establish up-to-date prevalence and incidence of diabetic retinopathy (DR) from diagnosis and in five-year intervals in a Swedish heterogeneous cohort, identify specific risk groups, correlate comorbidities and discern new biomarkers for DR.
Background: Altered life styles, changing demographics, increasing diabetes and new treatment modalities in Sweden in the 21st century demands up to date figures on prevalence and incidence of DR as well as identification of risk factors and treatment strategies. The regional program All New Diabetics in Skåne (ANDIS) gathers data from all newly diagnosed diabetics in all hospitals and primary care centers in Skåne. ANDIS-eye complications establishes DR incidence and prevalence in five-year intervals in a random sample of the ANDIS population.
Methods: Two thousand sixty-nine men (59.8%) and 1391 women (40.2%) were screened for DR according to the International Classification of Diabetic Retinopathy Severity Scale, during 2008–2013. Information on patient characteristics was collected from the ANDIS database. Statistic analysis was performed with SPSS 22 by one way-ANNOVA with *post hoc* Bonferroni correction and logistic regression.
Results: Mean age was 58.9 (± 13.6) years. The distribution of diabetes was 2.2% type 1, 92.2% type 2, 4.8% LADA, 0.1% MODY and 0.6% secondary diabetes. 72.8% of subjects originated from Sweden. At diagnosis 78.4% of subjects had no, 8.7% mild, 5.6% moderate, 0.4% severe and 0.1% proliferative DR. 1.4% had vision threatening DR. The prevalence was highest in type 2-diabetes (15.2%) and LADA (13.4%). The risk of DR was higher in men (59.7%) than in women (OR 1.3, $p = 0.011$), and in patients of 'non-Swedish origin', particularly Western Asian, than in Swedes ($OR = \geq 1.2$, $p < 0.001$).
Discussion: The overall prevalence of DR at diagnosis was low, which might reflect early diabetes diagnosis as well as DR screening. Risk factors for DR at diagnosis were, among others, male sex, non-Swedish origin, LADA and a high debut HbA1c. ANDISec envisages assessment of individual laboratory, medical, social and genetic risk factors in five-year intervals, to tailor individual screening and treatment options, as well to establish possible biomarkers for DR.

PP-UVE-090

The role of biologic drugs in the treatment of chronic pediatric uveitis at the Helsinki University Hospital

Sanna Leinonen, Päivi Lindahl, Kaisu Kotaniemi
Department of Pediatric Ophthalmology, Helsinki University Hospital, Helsinki, Finland

Aim of the Study: To evaluate the use and effect of biologic drugs in the treatment of JIA uveitis in Helsinki University Hospital 2006–2014.
Patients and Methods: The core of the patients consisted of 187 paediatric patients with chronic uveitis examined between 2006–2014 at the Helsinki University Hospital. Most of the patients had JIA diagnosed by a paediatric rheumatologist, none of the patients had a traumatic or infectious uveitis. The patients were examined regularly by an ophthalmologists.
Results: Altogether 70% of the 187 patients were treated with systemic medication. One third of the uveitis patients were treated with only DMARDs. Biologic drugs were given to 42% (78/187) of the patients.

The first line biologic drug combined with a DMARD was infliximab in 53%, adalimumab in 33% and etanercept or abatacept in 14% of the patients. Five patients were on biologic drug monotherapy because of compliance problems and a low number of the patients needed a low-dose systemic corticosteroid. The medication of 45% infliximab patients was changed to adalimumab and 11% of adalimumab treatments were changed to other biologic agents. In this data we also report of 4 patients who received golimumab as their biologic drug. In most of the patients the response to the biologic drugs was favourable. Main reasons for changing the biologic treatment were inefficacy and compliance problems.
Conclusion: In this study we describe the results of biologic treatment in chronic uveitis among paediatric patients at Helsinki University Hospital, Finland.

PP-VIT-010

Tobacco smoking is associated with advanced diabetic retinopathy in young adults with type 1 diabetes. The Oulu cohort study of diabetic retinopathy

Kaisu Järvinen, Mira Siiskonen, Nina Hautala, Aura Falck
Oulu University Hospital Department of Ophthalmology, Oulu, Finland

Purpose: To study the prevalence of preproliferative or proliferative diabetic retinopathy (advanced DR) in current and former smokers and non-smokers.
Methods: One hundred and twenty-three (72 men) 29 ± 3 year-old subjects with type 1 diabetes for 23 ± 4 years were examined for the stage of retinopathy, and reported their current or former smoking or non-smoking habits. They belonged to a population-based cohort dating 18 years back. 93 of 216 subjects of the original cohort failed to attend the examination and/or report smoking, or had died ($N = 8$).
Results: There were 59 non-smokers, 23 former and 41 current smokers. The prevalence of advanced DR was 29% in non-smokers, 48% in former smokers and 63% in current smokers. Of the 54 subjects with advanced DR, 48% were current and 20% former smokers, while 22% of the 69 subjects with less severe or no retinopathy were smoking and 17% had smoked before (p = 0.003). The association in male smokers was especially notable: 73% of the 26 currently smoking men had advanced DR.
Conclusions: Smoking is associated with advanced retinopathy. Even though another type of study design is needed to prove the causality, subjects with diabetes are suggested to be strongly advised not to start tobacco smoking.

PP-VIT-047

Cost effectiveness of ocriplasmin for the treatment of vitreomacular traction, including when associated with macular hole ≤ 400 μm: A Swedish perspective

Kerry Gairy[1], Craig Bennison[2], Stephanie Thurston[2], Benedicte Lescrauwaet[3], Jorgen Aaberg[1]
[1]*Alcon Nordic A/S, Copenhagen, Denmark,* [2]*Pharmerit Ltd, York, UK,* [3]*ThromboGenics NV, Leuven, Belgium*

Purpose: Persistent attachment between the vitreous and macula can cause vitreomacular traction (VMT) and subsequent formation of a macular hole (MH), resulting in reduced visual function. This analysis estimated long-term benefits and costs associated with treatment with ocriplasmin vs standard-of-care (watchful waiting followed by vitrectomy, if needed) from a Swedish societal perspective.
Methods: The model included (1) a short-term decision-tree, simulating 6-month anatomical and visual outcomes observed in the two Phase III MIVI-TRUST ocriplasmin trials (006/007) and (2) a long-term Markov state-transition model, following the patient cohort over a lifetime-period. The short and long-term models were linked through common health states based on VMT resolution, number of vitrectomies and visual acuity (VA) status. Effectiveness and safety outcomes were based on the MIVI-TRUST, randomized, double-masked, placebo-controlled trials. Benefit was measured in quality-adjusted life years (QALYs), estimated based on (1) time spent in VA health states defined by best- and worst-seeing eye, (2) impact of metamorphopsia (3) impact of surgical interventions (vitrectomy and phacoemulsification) and (4) impact of treatment-related adverse events. Resource use was based on Swedish hospital clinical guidelines. Unit costs were drawn from published national and regional sources in Sweden.
Results: The additional cost of treatment with ocriplasmin was partially offset by cost savings and quality of life benefit from the avoidance of surgical procedures, adverse events and blindness. Over a life-time, ocriplasmin generated incremental benefits in terms of QALYs and additional costs of 0.094 and 27 286 SEK, respectively, over standard of care. The incremental cost effectiveness ratio (ICER) was 289 980 SEK.
Conclusions: Ocriplasmin represents a cost-effective use of Swedish healthcare resources in patients with VMT, including when associated with MH ≤ 400 μm.

PP-VIT-055

Retinal detachment from CSF leakage in Morning Glory syndrome

Panagiotis Salvanos[1,2], Erlend Landsend[1], Ralf Øvergaard[1]
[1]*Department of Ophthalmology, Oslo University Hospital, Oslo, Norway,* [2]*University of Oslo, Oslo, Norway*

Aim: To present a child with Morning Glory syndrome and retinal detachment from leakage of cerebrospinal fluid (CSF).
Material & Methods: A 5-year-old male patient was referred to the Department of Ophthalmology, Oslo University Hospital due to chronic retinal detachment from Morning Glory syndrome. The patient was operated with pars plana vitrectomy, peripapillary laser photopexy and silicone oil injection. A sample of the subretinal fluid taken perioperatively (0.5 ml) was analyzed for the presence of x-beta trace protein (sensitive and specific marker of CSF).
Results: The result of the examination was strongly positive for the presence of x-beta trace protein, confirming that the CSF was the origin of the subretinal fluid. The retina remained attached during the postoperative period, except for the inferior peripapillary area.
Conclusion: Discovery of x-beta trace protein in the subretinal fluid gives more insight in the pathophysiology of retinal detachment in patients with Morning glory syndrome.

S02

A critical review of: 'Early SD-OCT diagnosis followed by promt treatment of radiation maculopathy using intravitreal bevacizumab maintains functional visual acuity', 'Combination therapy with triamcinolone actonide and bevacizumab for the treatment of severe radiation maculopathy in patients with posterior uveal melanoma' Shah N *et al*, Clin Ophthalmology 2012:6 1739–1748 and 2013:7 1877–1882

Nils Eide
Eye Department, Oslo University Hospital, Oslo, Norway

Purpose: To report criteria for treatment and document results from radiation maculopathy.
Methods: An evaluation of two articles with material from 2006 to 2009, that included 159 patients treated with iodine-125 plaque for intraocular melanoma.

Spectral domain optical coherence tomography (SD-OCT) was used to evaluate the macula using a 1–6 gradation. Treatment was started when examination revealed signs of macular oedema. Bevacizumab 1.25 mg/0.05 ml was injected intravitreally. Treatment with triamcinolone 4 mg/0.1 ml was added when the condition was refractory or the oedema worsened.

Tumours were treated with 85 Gy to the apex. Patients were evaluated at 2–4 month intervals. 159 cases were included. 25 of them were evaluated after receiving a combination treatment. They had received either 3 or more injections with minimal improvement or the maculopathy was defined as OCT grade 5 or 6 oedema.

Results: On average, OCT showed the first sign of oedema by 12 months. 51% (81/159) of the patients treated had 20/50 or better vision at median follow-up of 36 months. The patients were given a mean of 5 injections (1–17) over a mean of 18 months (1–54), starting at 16 months (3–30) after plaque therapy.

A subset of the patients developed severe maculopathy refractory to anti-VEGF monotherapy. 36% of them demonstrated 20/50 or better vision at last follow-up.

Conclusion: Radiation maculopathy is reported in over 50% of the eyes after plaque radiotherapy. Unfortunately, neither the total number of tumours irradiated, nor the number affected by radiation maculopathy, is given.

The authors have made a consecutive, retrospective study, showing that a systematic OCT evaluation with a treatment algorithm can be used to preserve vision after irradiation. This follow-up regime will require more resources. OCT detected radiation side effects already at 4 months. As documented in this study, patients' vision after three years may benefit from this practice.

S16

Retinal volume changes after macular hole surgery

Juris Vanags, Guna Laganovska
P. Stradins Clinical University hospital, Riga, Latvia

Purpose: Internal limiting membrane peeling is standard procedure for macular hole surgery. The procedure may cause traumatic changes in retinal structure, leading to atrophic changes in neuroretinal layer. Aim was to find out retinal volume changes after macular hole surgery, using 2 different ILM peeling techniques.

Methods: Retrospective analysis of 33 macular hole surgery patients (33 eyes), surgery performed in 2013, was done. ILM peeling was performed by means of diamond dusted sweeper or by means of ILM forceps. SDOCT (Heidelberg Spectralis) was performed before surgery and in follow-up visits (3 or 6 months), using preset Dense Volume scan (49 lines, ART 16, 20 degrees). Standard Volume analysis of 1.3 and 6 mm in diameter was performed. Retinal volume changes for potential atrophy were analyzed in peripheral 3–6 mm subfields, as central subfields changes anyway due to closure of macular hole.

Results: ILM peeling by means of diamond dusted sweeper for 13 eyes was done (group 1), by means of ILM forceps- 20 eyes (group 2). Mean retinal volume in superior subfield of group 1 before surgery was 1.5392 mkl, after – 1.5315 mkl ($p < 0.05$), in nasal subfield- 1.5831 mkl and 1.5834, inferior- 1.4669 mkl and 1.4600 mkl ($p < 0.05$), temporal- 1.4623 mkl and 1.4392 ($p < 0.05$). Retinal volume changes in group 2 were- superior subfield- 1.5930 mkl and 1.5940 mkl, nasal- 1.7420 and 1.7415, inferior- 1.5980 and 1.5940, temporal- 1.6510 and 1.6503, no statistical significance found.

Conclusions: ILM peeling by means of ILM forceps (group 2) appears to be less traumatic compared to ILM peeling by means of diamond dusted sweeper (group 1) as significant retinal volume decrease occurs only in group 1.

S16

Influence of the vitreomacular interface on anti-VEGF therapy in wet-AMD patients after cataract surgery

Liga Radecka, Eriks Elksnis, Ilze Lace, Guna Laganovska
Riga Stradins university, Riga, Latvia

Aim: To determine whether posterior vitreoretinal attachment/ posterior vitreous detachment play any role in reactivation of wet-AMD after cataract surgery.

Materials and Methods: Retrospective study of Optical Coherence Tomography data and medical records of 48 patients with treated wet-AMD who undergone cataract surgery. All patients received repeated intravitreal injections with Bevacizumab. After the stabilisation for at least 3 months period cataract surgery was done. According to the presence of posterior vitreous detachment prior to the cataract surgery (documented by OCT) all patients were categorised into 2 groups (1st group – eyes with posterior vitreous detachment, 2nd group – eyes with posterior vitreoretinal attachment). Central retinal thickness (CRT) changes 1 month after cataract surgery compared to data prior to the treatment and before the surgery were analyzed in both groups.

Results: In 18 eyes (37.5%) there were posterior vitreous detachment, but in 30 eyes (62.5%) there were posterior vitreoretinal attachment prior to cataract surgery.

In posterior vitreous detachment group mean CRT before wet-AMD treatment was 485.26 μm, prior to the cataract surgery – 353.61 μm, but 1 month after cataract surgery it was 339.26 μm. Mean difference in CRT was reduction of 146 μm or 30.09% ($p < 0.01$).

In posterior vitreoretinal attachment group mean CRT before wet-AMD treatment was 438.00 μm, before the cataract surgery – 359.83 μm and 1 month after cataract surgery – 361.46 μm (SD 98.70 μm). Mean difference in CRT was reduction of 76.54 μm or 17.47% ($p < 0.01$).

In patients with posterior vitreoretinal attachment central retinal thickness changes slightly if compare CRT before treatment and 1 month after surgery ($p = 0.01$). But in group with posterior vitreous detachment CRT changes average ($p = 0.01$).

Conclusions: Our study confirms that in patients with posterior vitreous detachment central retinal thickness before treatment and 1 month after cataract surgery changes more than in posterior vitreoretinal attachment group.

S16

Late results of retinal detachment surgery with 360 degree retinotomy

Guna Laganovska[1,2], Liga Radecka[1,2], Juris Vanags[1,2]
[1]*Riga Stradins University, Riga, Latvia,* [2]*P. Stradins University Hospital, Riga, Latvia*

Aim: to analyze postoperative results five years after retinal detachment surgery with 360 degree retinotomy.

Material and Methods: Retrospective analysis of 11 patients medical records operated during 2008 were analysed. We analyse visual aquity, retinal attachment, and secondary glaucoma five year after surgery.

Results: Age of patients 16–75 years. 8 – males, 3 females. In all cases pars plana vitrectomy with 360 degree retinotomy and siliconoil tamponade were performed. All patients were aphakic. In 7 cases reason for retinal detachment were rhegmatogenus detachment, in 4 cases trauma. In 2 cases 360 degree retinotomy was performed in first surgery, in 7 cases – in second surgery, in 2 cases in 3 th surgery. Visual aquity 1 month after surgery were 0.02 (range hand movement – 0.06), five years after surgery 0.04 (light perception – 0.1). One month after surgery retina were attached in all cases, five years after surgery in 4 cases retinal detachment were found. One month after surgery secondary glaucoma were in 3 cases, five years after surgery in 6 cases.

Conclusions: Retinal detachment surgery with 360 degree retinotomy is one of the possibilities for vision recovery in difficult cases of retinal detachment.

S21

AMD – risk factors and possibilities for prevention

Inger Christine Munch[1,2]

[1]Department of Ophthalmology, Roskilde Hospital, Roskilde, Denmark, [2]Faculty of Health Sciences, University of Copehagen, Copenhagen, Denmark

Currently, age-related macular degeneration (AMD) is untreatable, except for its neovascular complications, where pharmaceutical inhibitors of vascular endothelial growth factor (VEGF) have revolutionized the management and prognosis for this condition. The age-related eye disease study (AREDS) supplements reduce the risk of developing neovascular AMD but has no effect on the disease *per se* or on the risk of geographic atrophy, even with the AREDS 2 formula having replaced beta-carotene by lutein and zeaxanthin.

Preventive measures target smoking, obesity, diet and exercise. We have recently shown that being active 7 hr per week reduces the odds for very early AMD (drusen > 63 μm) by 67%. Updated recommendations for clinical counselling and risk management will be presented.

S22

Cataract surgery after penetrating keratoplasty

Kari Krootila, Waldir Neira

Helsinki University Eye Hospital, Helsinki, Finland

Purpose: To evaluate the postoperative results after implantation of toric lens in patients with cataract and history of prior penetrating keratoplasty (PKP)

Methods: This retrospective study includes a total of 18 post-PKP patients with cataract and moderate to high astigmatism that underwent phacoemulsification and toric lens implantation [SN60T series, Alcon, Fort Worth, TX, SN60T5, SN60T7, SN60T9 (6, 2 and 10 eyes, respectively)]. Postoperatively, best spectacle corrected visual acuity (BCVA), and manifest refraction were measured.

Results: All patients showed improvement of BCVA postoperatively (BCVA logMAR pre 0.73 ± 0.32 vs post 0.34 ± 0.15). Preoperatively topographic astigmatism was 4.64 ± 2.85 diopters (D). Postoperatively, manifest refraction cylinder was 2.65 ± 1.62 D. Six and 11 patients achieved ± 1.0 D and ± 2.0 D of intended correction, respectively. Postoperative cylinder values were closer to the intended in lower corrections.

Conclusions: Use of toric intraocular lens during cataract surgery patients with prior PKP seems to reduce the manifest refraction cylinder, and improve the BCVA. Yet, postoperative results showed a wide range of achieved cylinder correction.

S22

Management of postkeratoplasty astigmatism in keratoconus patients

Kari Krootila, Olli Wetterstrand, Juha Holopainen

Helsinki University Eye Hospital, Helsinki, Finland

Purpose: Femtosecond (FS)-assisted intrastromal relaxing incisions (FS-ISRI) are effective option to treat high levels of astigmatism following penetrating keratoplasty (PK). Here we have retrospectively evaluated the efficacy of FS-ISRI and correlated the efficacy with the posterior depth of the incision.

Setting: Helsinki University Eye Hospital, Helsinki, Finland.

Methods: Twenty eyes of 20 patients were treated for regular astigmatism after PK. Regarding the refractive status all eyes had stable refraction and sutures had been removed previously. Corneal thickness was measured prior to FS-ISRI with ultrasound pachymetry to calculate the incisional depth. FS-laser assisted paired arcuate incisions were made inside the graft stroma to an anterior depth of 90 μm thus preserving the epithelium. Ten eyes were operated with a 125 μm marginal of the posterior intact cornea and 10 eyes with a 90% corneal depth (i.e. 10% of the posterior corneal depth was left unoperated). Clinical follow-up consisted of biomicroscopy, intraocular pressure measurement, fundus examination and topographic evaluation using anterior segment OCT (Tomey SS-1000 Casia) at one and 3 months. Corneal thickness and postoperative depth of incisions was measured from OCT scans.

Results: Best corrected visual acuity (BCVA) improved in the 90% group ($p < 0.05$) but remained constant in the 125 μm group. Refractive cylinder decreased in the 90% group but remained unchanged in the 125 μm group. Topographic anterior cylinder decreased by 38.5 ± 40.0% and 15.9 ± 16.8% respectively ($p < 0.05$). The smaller the posterior intact corneal margin was, the higher the surgically induced astigmatism was (SIA, $p < 0.05$). One patient had a graft rejection episode that was successfully treated.

Conclusions: FS-laser assisted intrastromal relaxing incisions are effective treatment for post-PK astigmatism. The efficacy seems to improve with increasing the posterior depth of the arcuate incisions.